

Big Bend Seagrasses Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

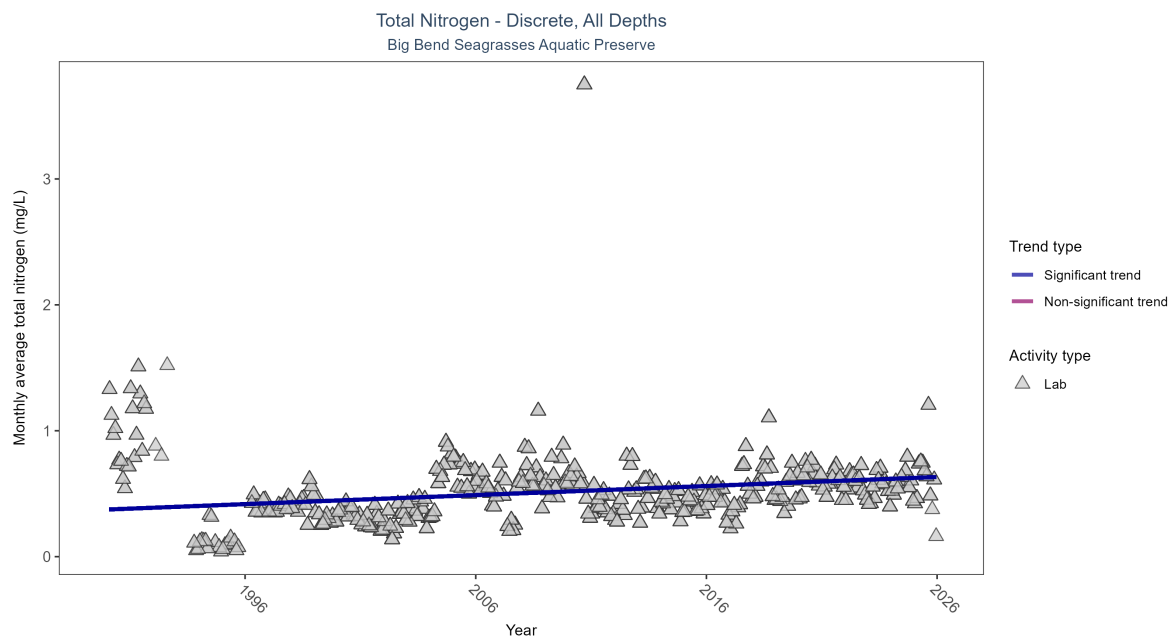


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	8320	36	1990 - 2025	0.466	0.22792	0.37408	0.00718	0

Monthly average total nitrogen increased by 0.01 mg/L per year.

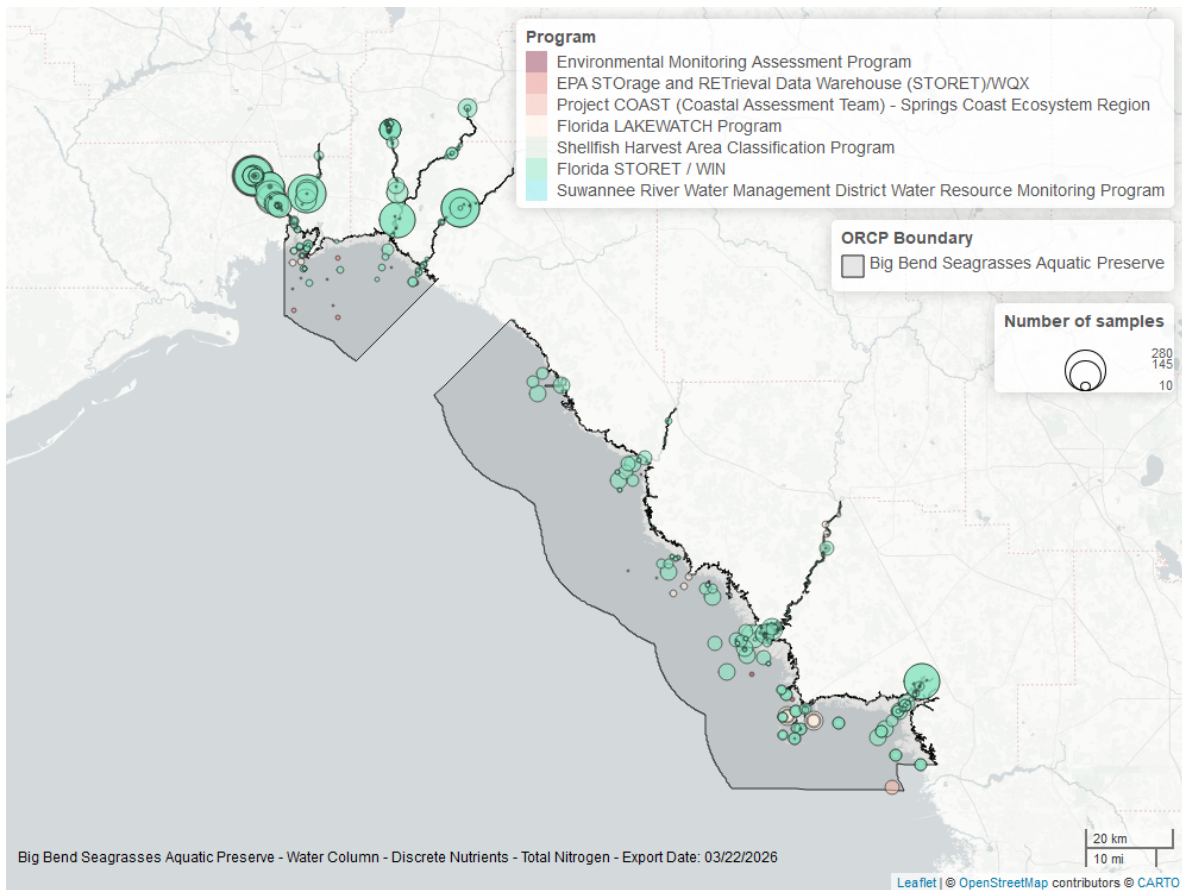


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

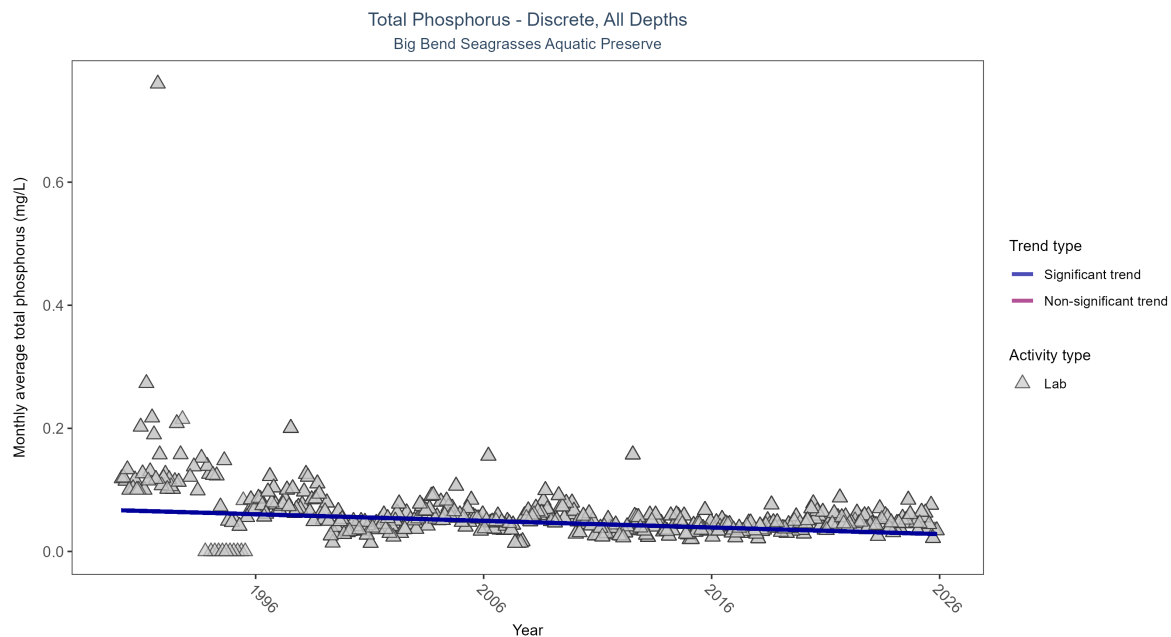


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	8781	36	1990 - 2025	0.042	-0.28848	0.06692	-0.00108	0

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

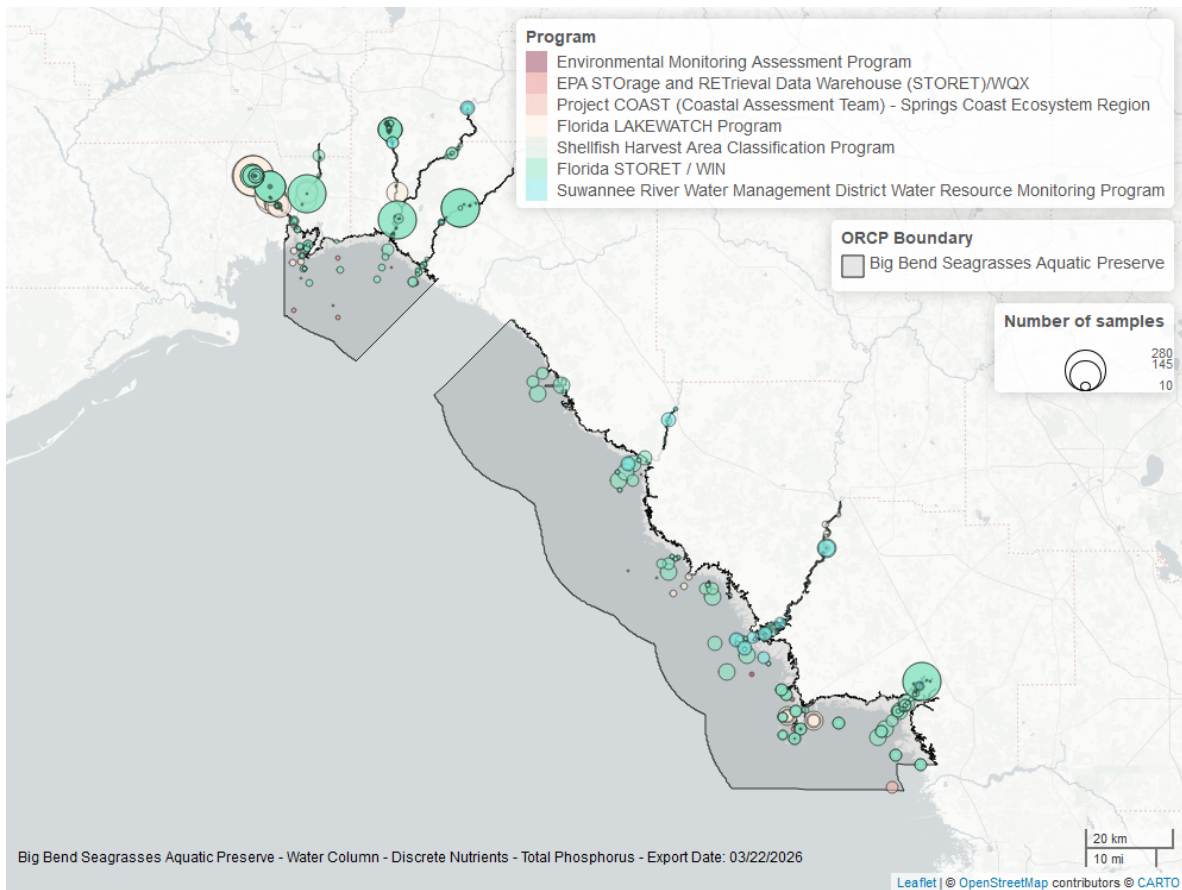


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

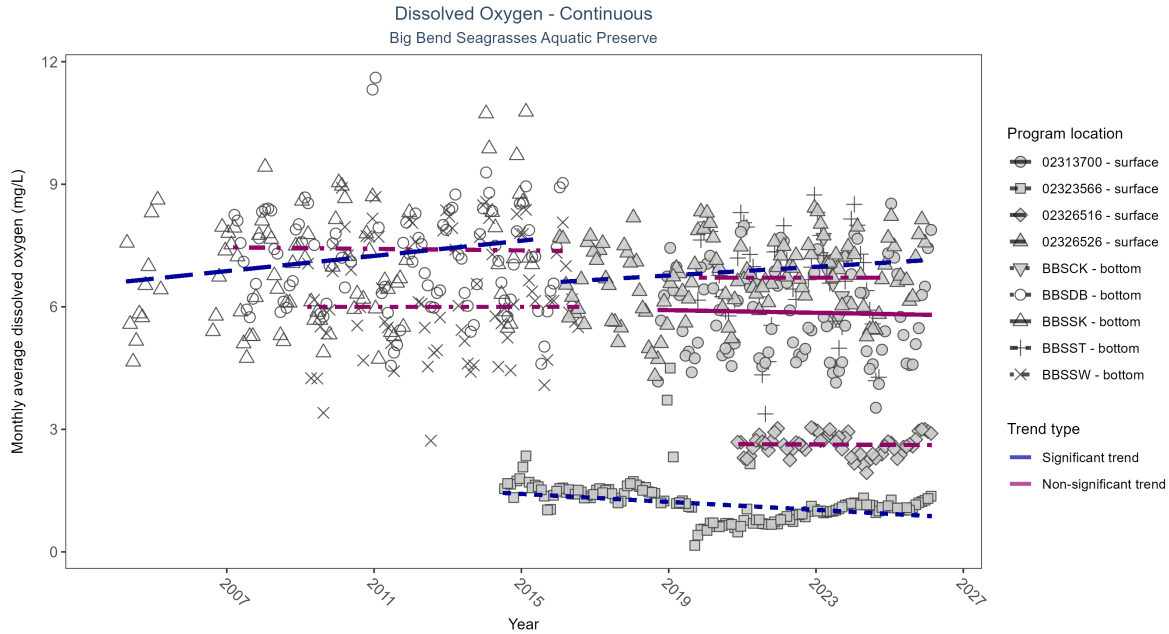


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	No significant trend	2604	9	2018 - 2026	5.6	-0.09	5.93	-0.02	0.3834
02323566	Significantly decreasing trend	4059	13	2014 - 2026	1.2	-0.33	1.47	-0.05	0
02326516	No significant trend	1764	7	2020 - 2026	2.6	-0.03	2.64	0	0.947
02326526	Significantly increasing trend	3468	10	2016 - 2025	6.7	0.18	6.61	0.05	0.0123
BBSCCK	Insufficient data to calculate trend	14788	1	2023 - 2023	6.7	-	-	-	-
BBSDDB	No significant trend	184327	10	2007 - 2016	7.3	-0.05	7.46	-0.01	0.6066
BBSSK	Significantly increasing trend	134287	10	2004 - 2015	7.1	0.28	6.6	0.09	0.0023
BBSSST	No significant trend	146149	6	2019 - 2024	6.9	-0.01	6.71	0	1
BBSSW	No significant trend	182327	8	2009 - 2016	6.2	0	6	0	1

At two program locations, monthly average dissolved oxygen increased by 0.05 mg/L per year at one site and by 0.09 mg/L per year at the other. At one program location, monthly average dissolved oxygen decreased by 0.05 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at five locations. There was insufficient data to fit a model for one location.

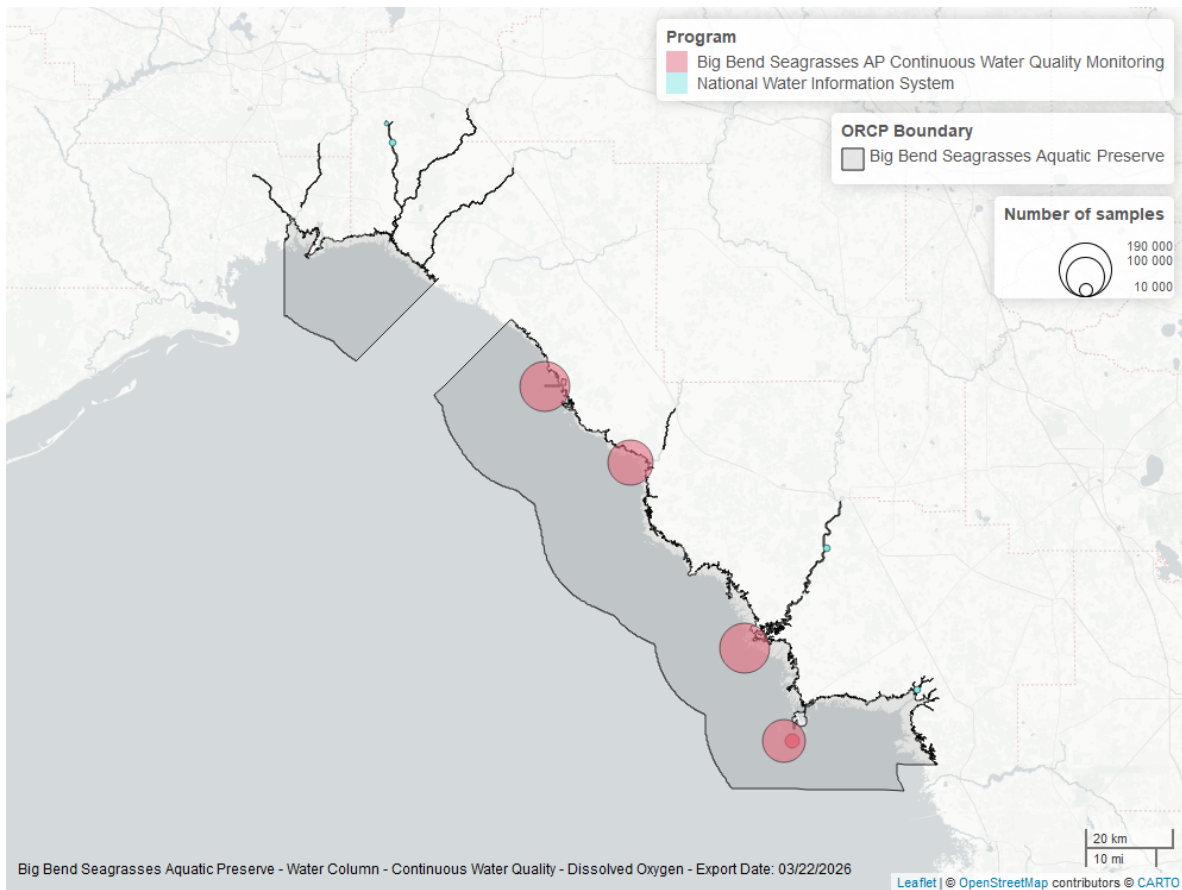


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

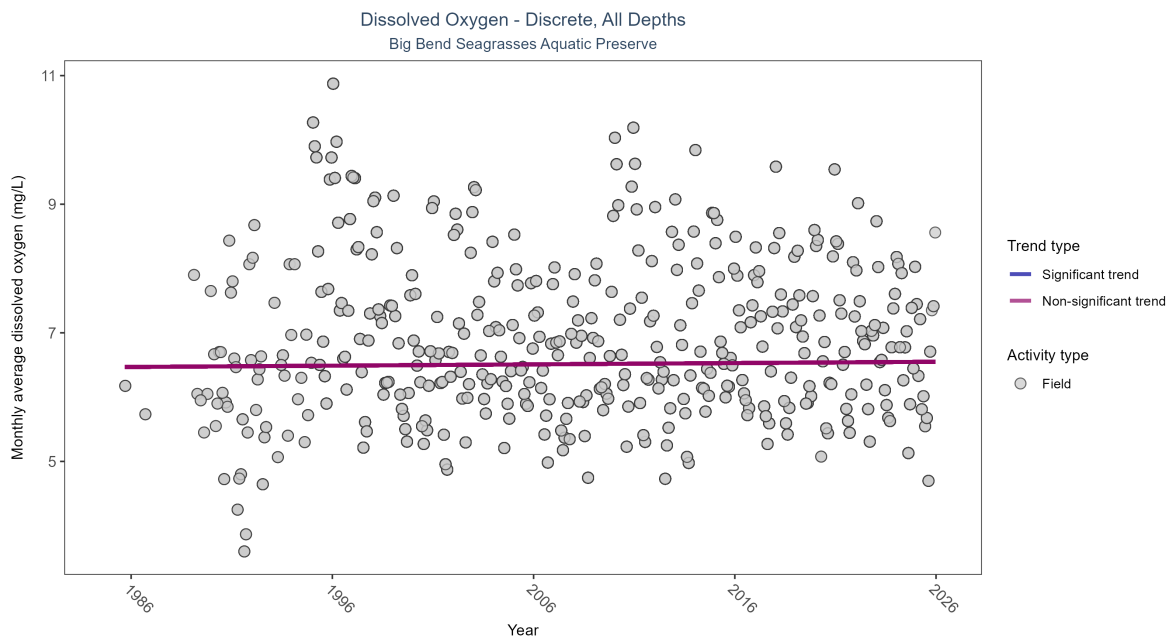


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	153302	39	1985 - 2025	6.7	0.0145	6.46672	0.00202	0.632

Dissolved oxygen showed no detectable trend between 1985 and 2025.

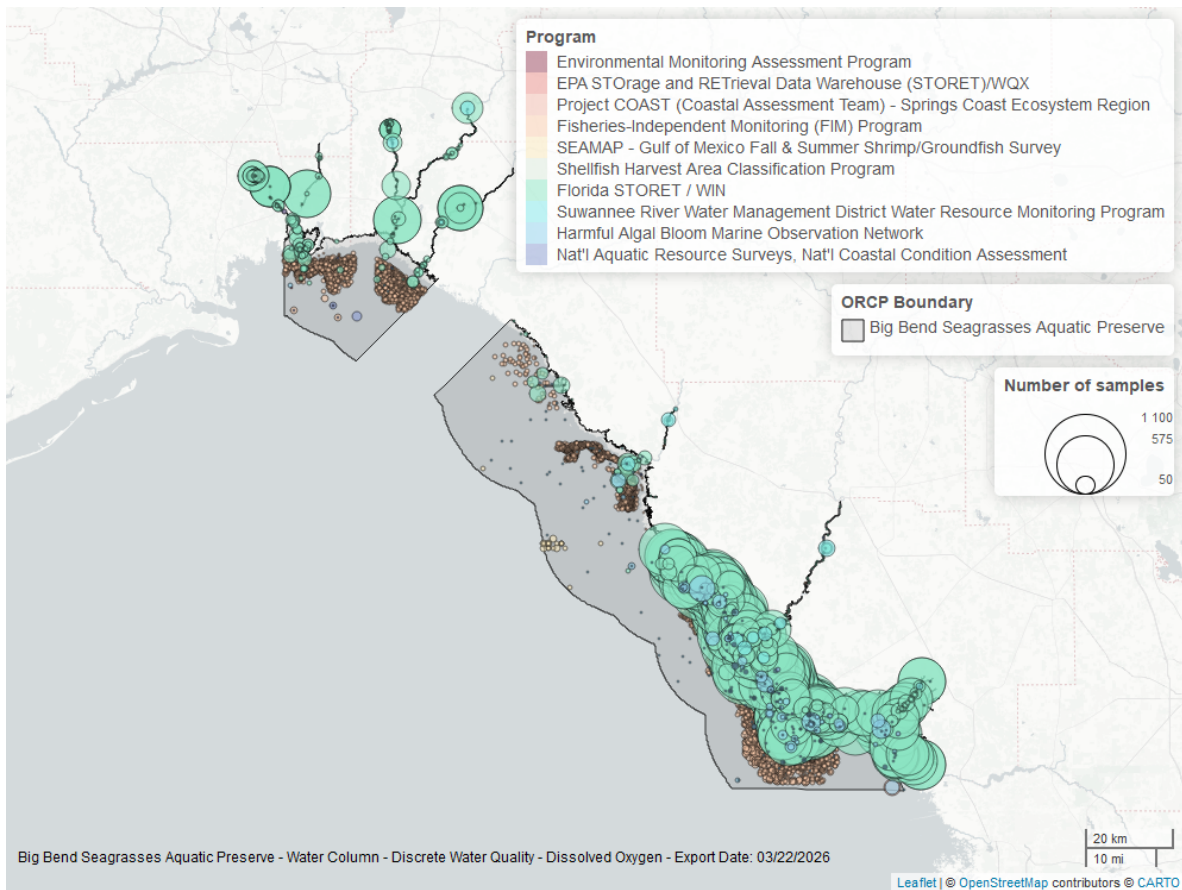


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

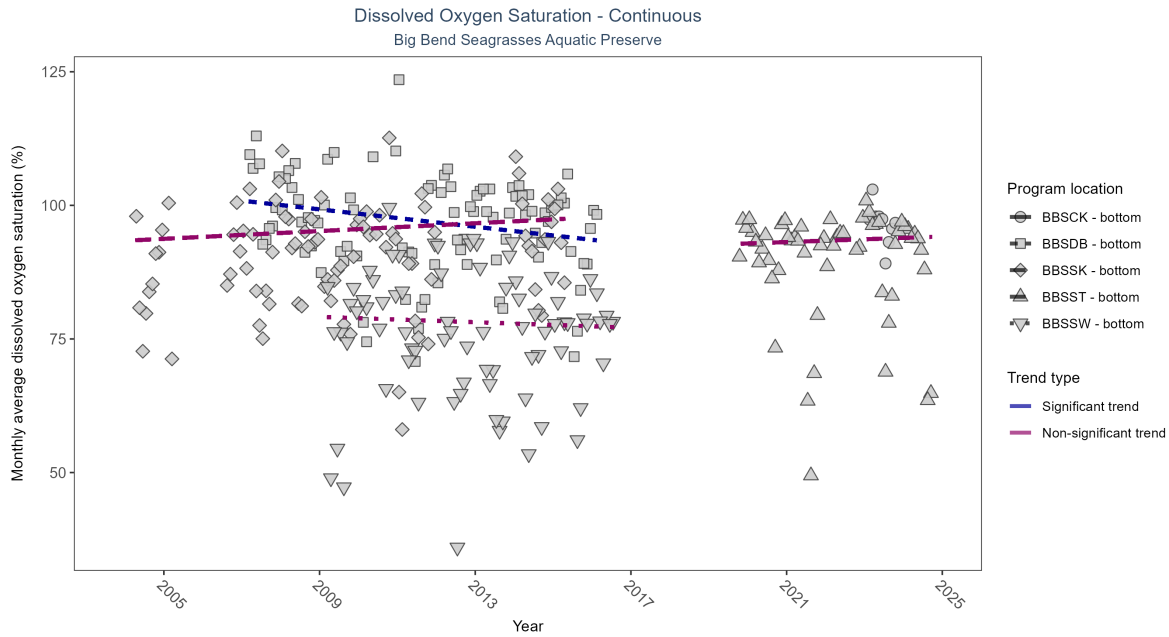


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSC	Insufficient data to calculate trend	19834	1	2023 - 2023	95.7	-	-	-	-
BBSD	Significantly decreasing trend	183530	10	2007 - 2016	97.6	-0.3	100.88	-0.81	0.0003
BBSS	No significant trend	134196	10	2004 - 2015	91.8	0.12	93.37	0.36	0.2612
BBSST	No significant trend	149203	6	2019 - 2024	93.3	0.08	92.61	0.26	0.4975
BBSW	No significant trend	182158	8	2009 - 2016	75.8	-0.05	79.14	-0.25	0.6835

At one program location, monthly average dissolved oxygen saturation decreased by 0.81% per year. No detectable change in monthly average dissolved oxygen saturation was observed at three locations. There was insufficient data to fit a model for one location.

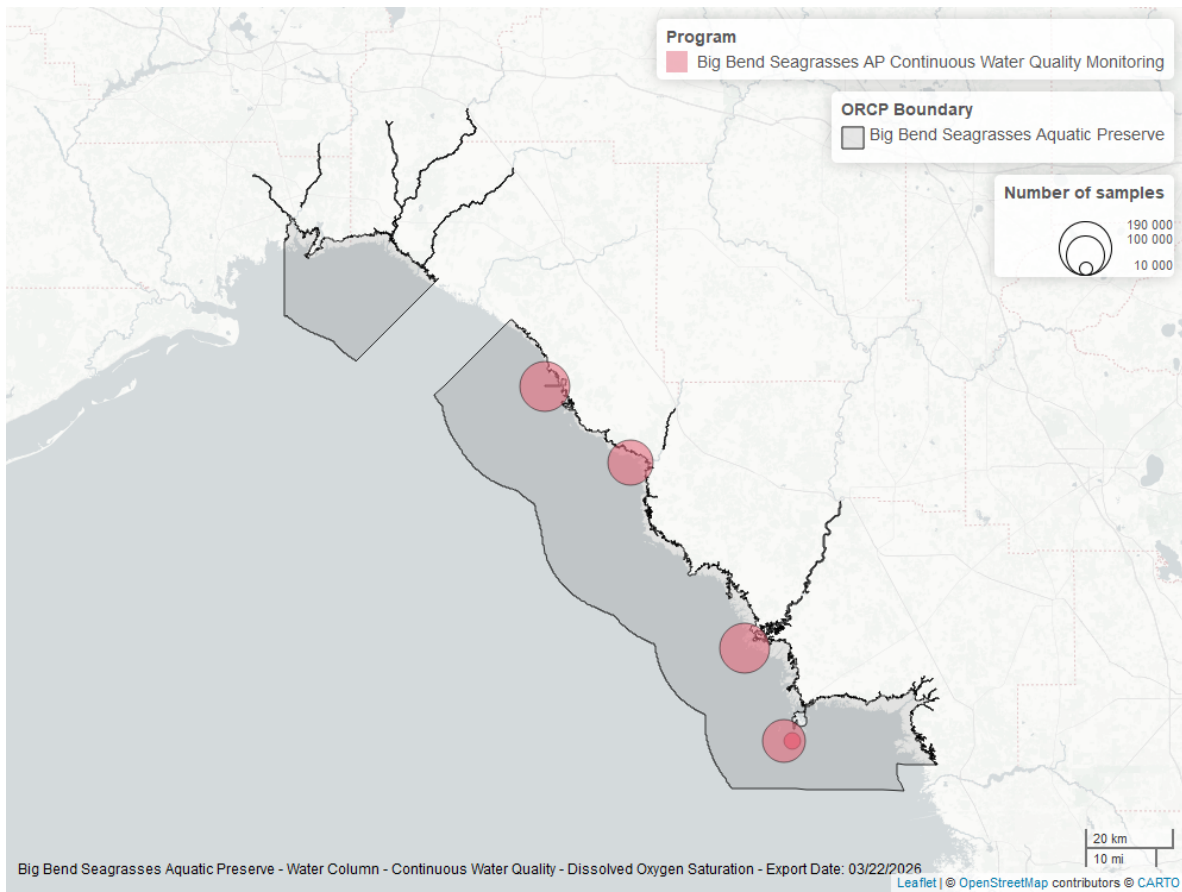


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

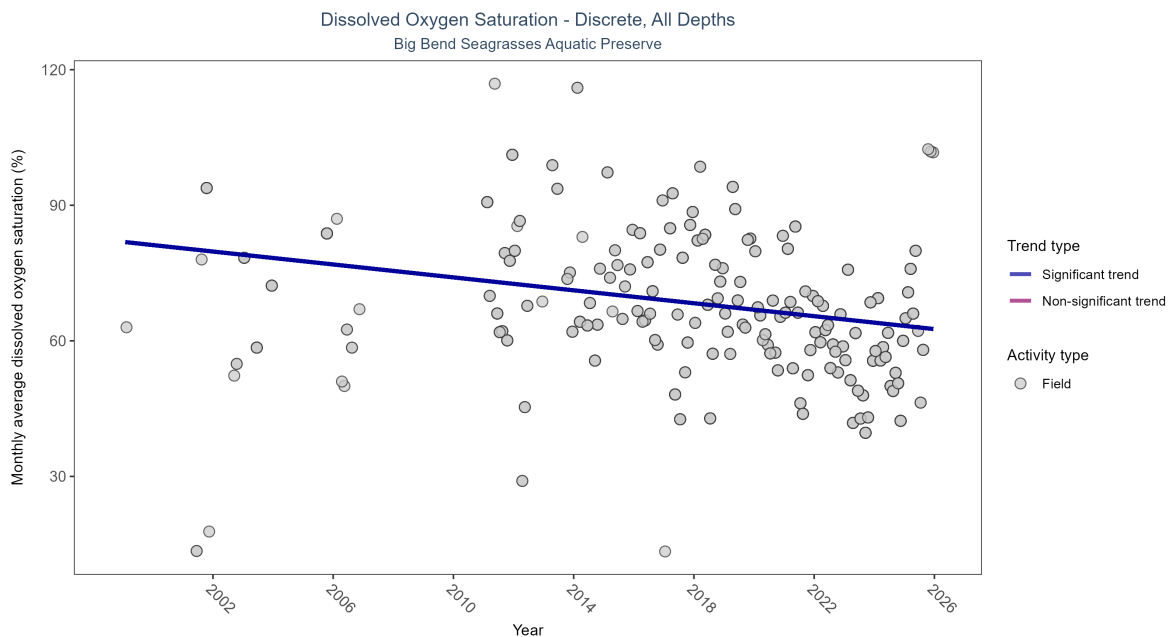


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	1987	21	1999 - 2025	67.1	-0.18751	81.90922	-0.71523	0.0018

Monthly average dissolved oxygen saturation decreased by 0.72% per year.

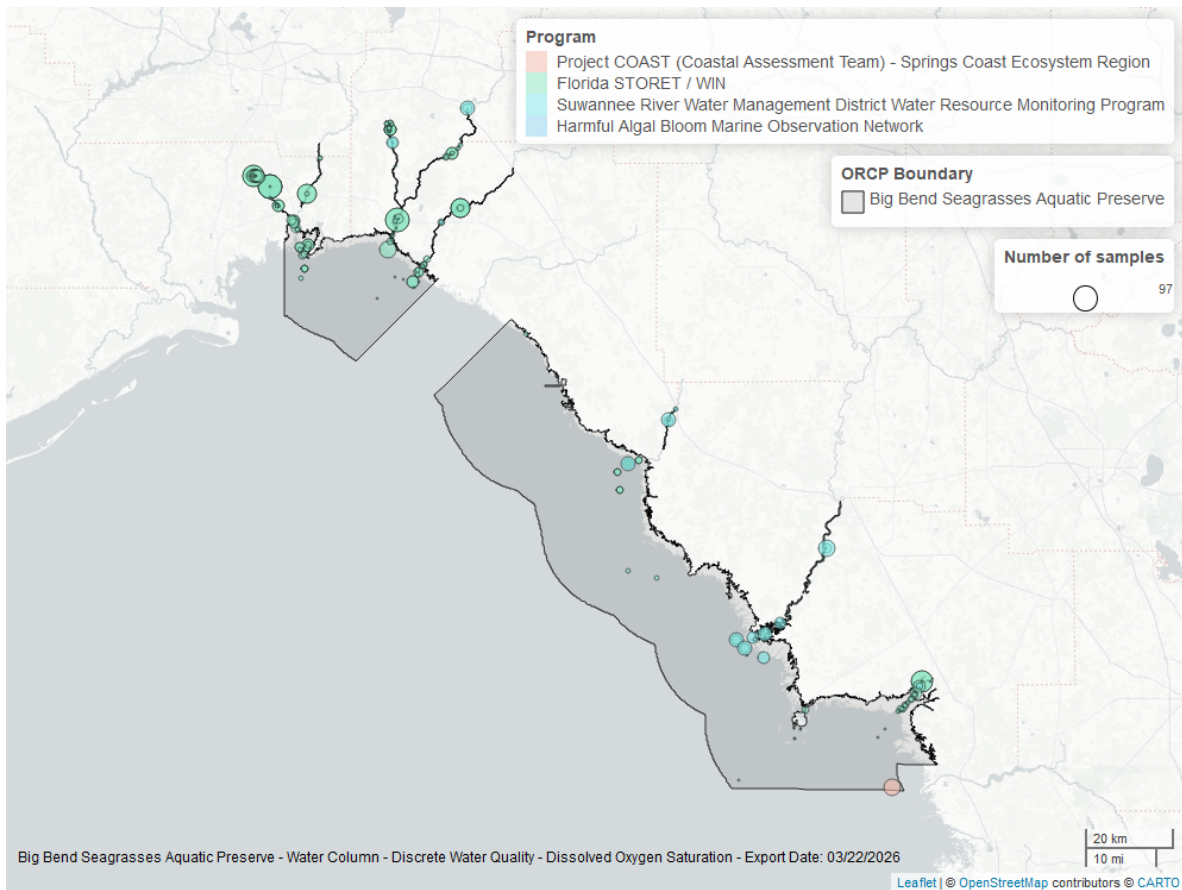


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

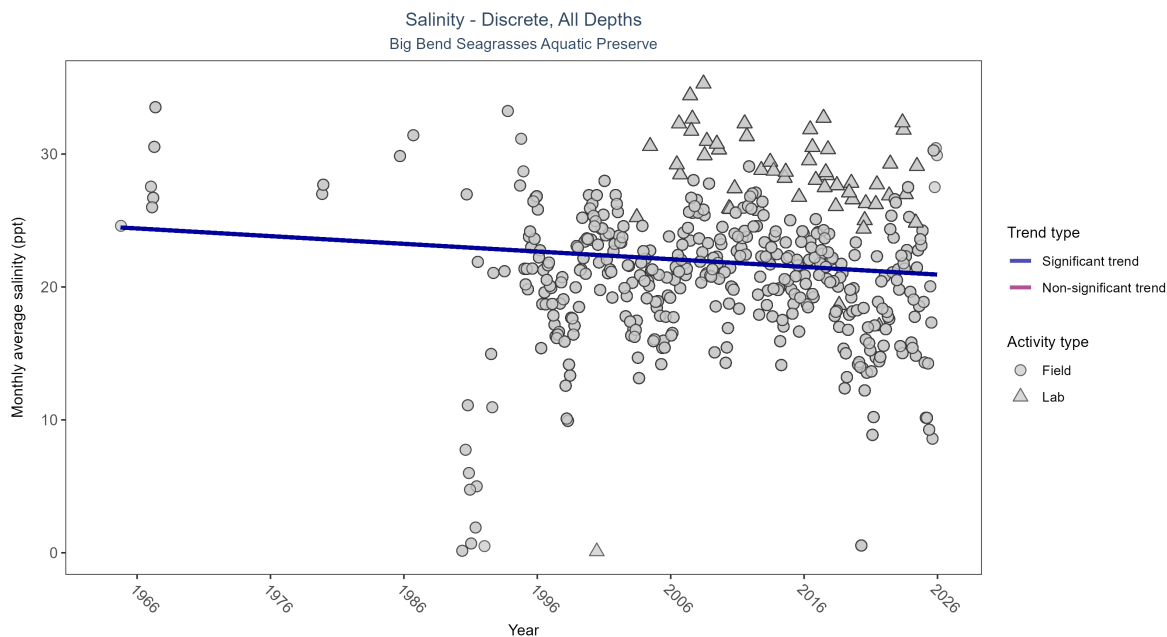


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly decreasing trend	158440	41	1964 - 2025	23.3	-0.0998	24.51876	-0.05786	0.0049

Monthly average salinity decreased by 0.06 ppt per year.

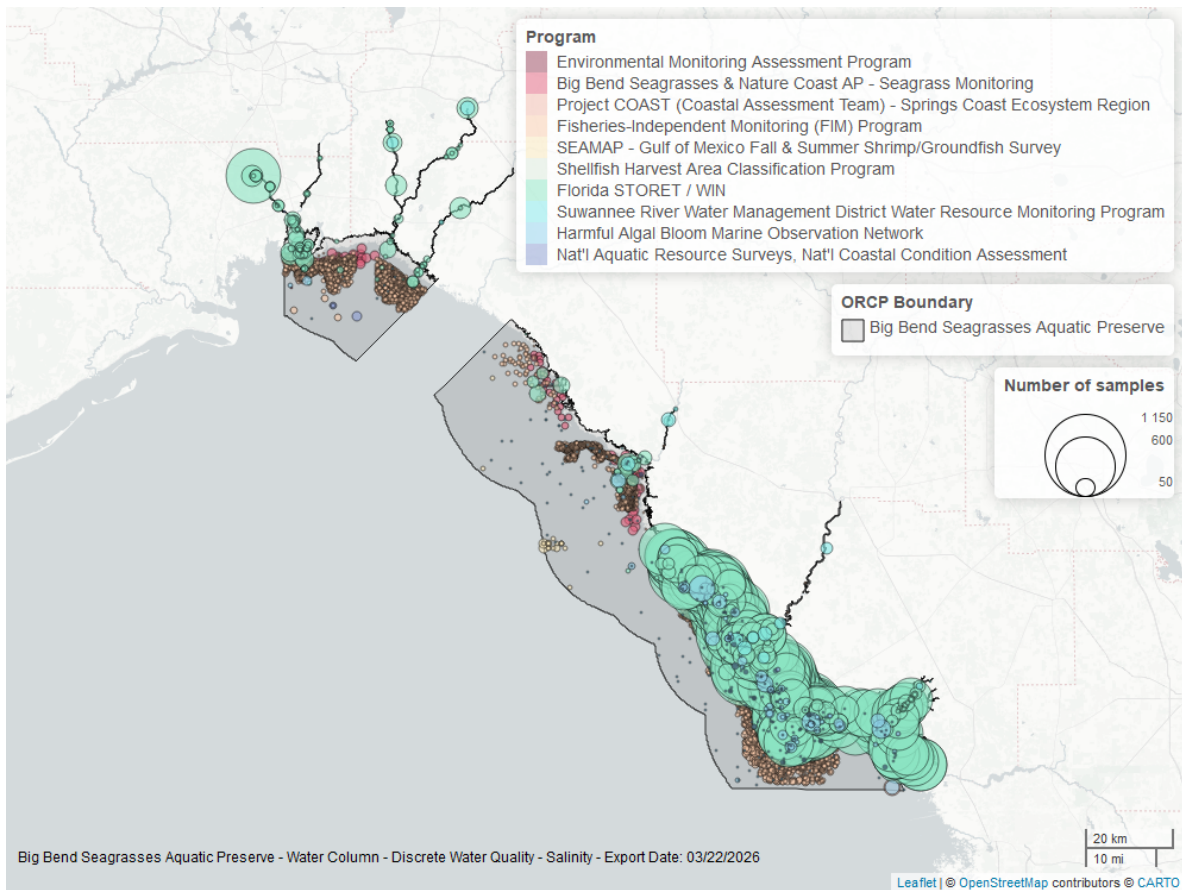


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

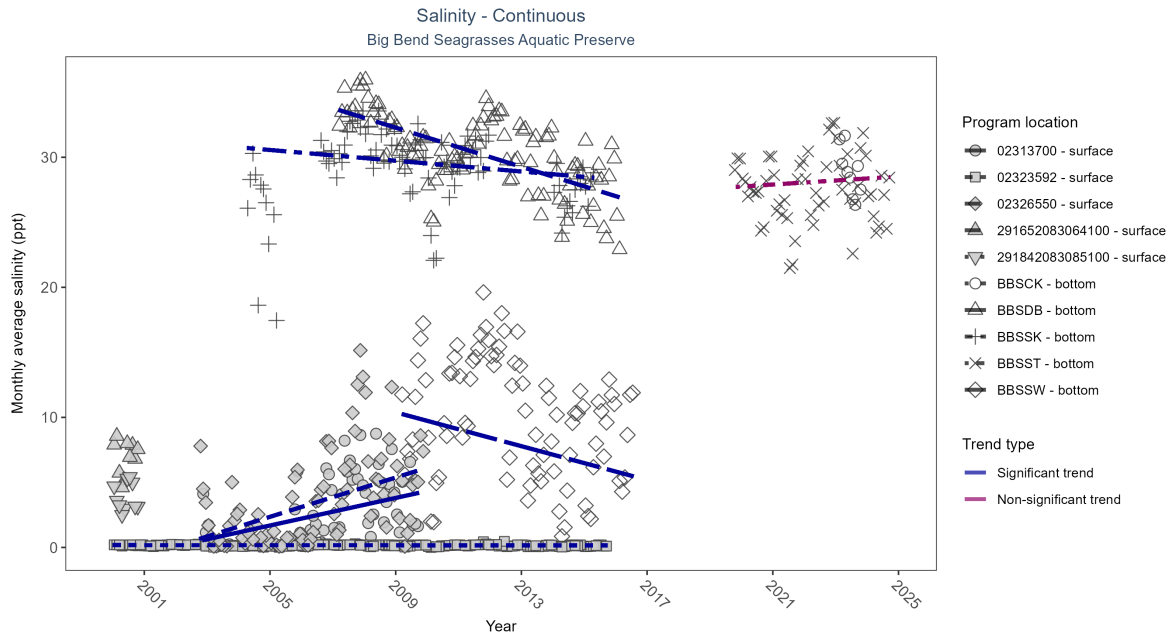


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Significantly increasing trend	1601	7	2002 - 2009	2.1	0.52	0.1	0.53	0
02323592	Significantly decreasing trend	6064	16	2000 - 2015	0.1	-0.13	0.18	0	0.0156
02326550	Significantly increasing trend	2507	8	2002 - 2009	1.7	0.48	0.09	0.76	0
291652083064100	Insufficient data to calculate trend	827	1	2000 - 2000	6.7	-	-	-	-
291842083085100	Insufficient data to calculate trend	584	1	2000 - 2000	3.7	-	-	-	-
BBSCCK	Insufficient data to calculate trend	14782	1	2023 - 2023	29.2	-	-	-	-
BBSDDB	Significantly decreasing trend	265544	10	2007 - 2016	30.6	-0.53	33.78	-0.75	0
BBSSK	Significantly decreasing trend	178356	10	2004 - 2015	29.6	-0.22	30.77	-0.21	0.0197
BBSSST	No significant trend	156720	6	2019 - 2024	28.5	0.08	27.59	0.15	0.516
BBSSW	Significantly decreasing trend	221696	8	2009 - 2016	7.5	-0.23	10.39	-0.65	0.016

At two program locations, monthly average salinity increased by 0.53 ppt per year at one site and by 0.76 ppt per year at the other. At four program locations, monthly average salinity decreased between less than 0.01 and 0.75 ppt per year. No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for three locations.

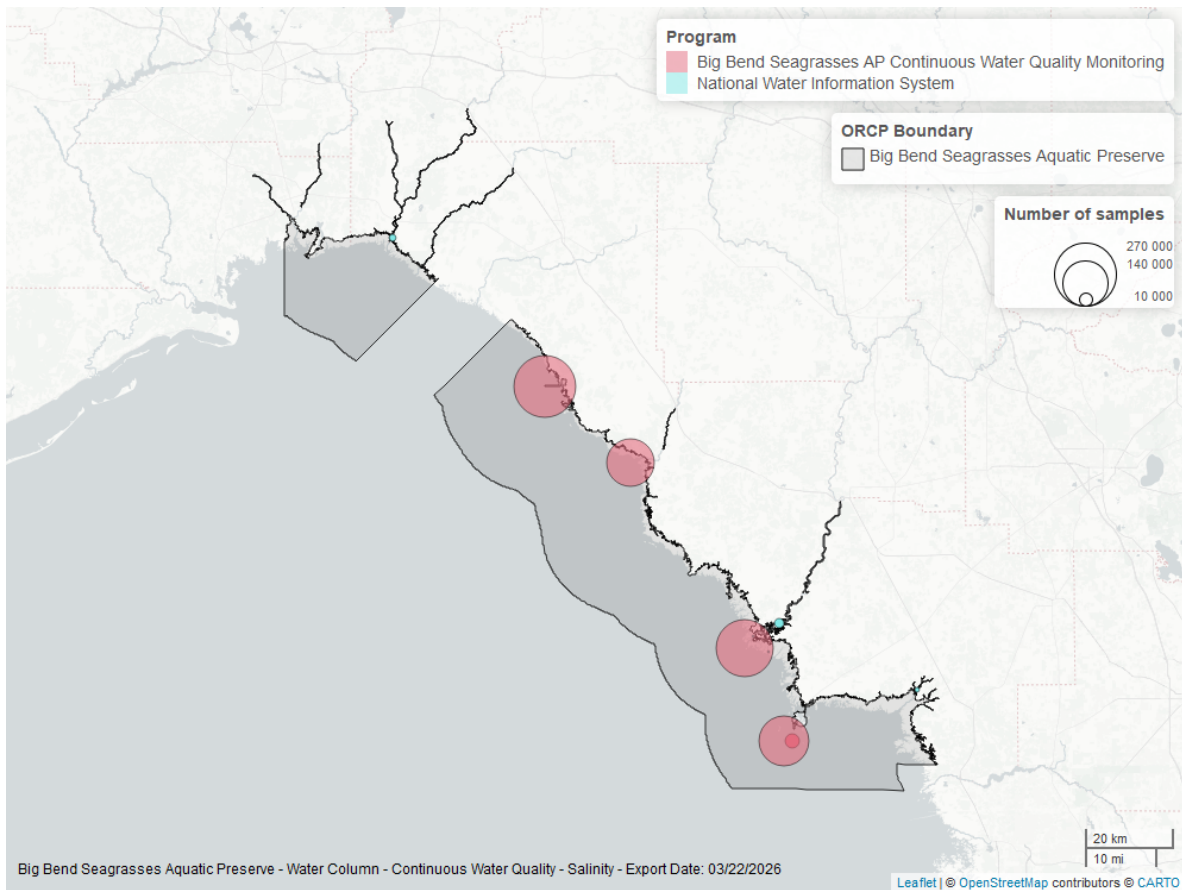


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

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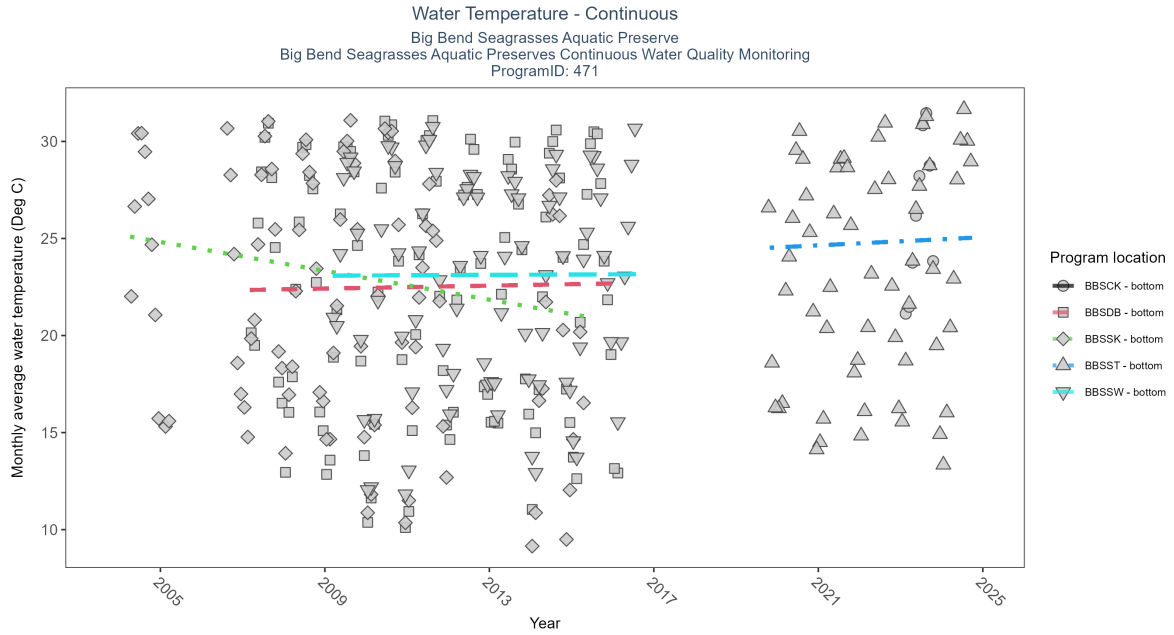


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results for All Stations - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCCK	Insufficient data to calculate trend	22232	1	2023 - 2023	26.4	-	-	-	-
BBSDDB	No significant trend	265988	10	2007 - 2016	23.2	0.08	22.34	0.04	0.2922
BBSSK	Significantly decreasing trend	179213	10	2004 - 2015	21.7	-0.33	25.19	-0.37	0.0002
BBSSST	No significant trend	163227	6	2019 - 2024	23.5	0.1	24.44	0.1	0.516
BBSSW	No significant trend	227996	8	2009 - 2016	23.7	0.02	23.09	0.01	0.9025

At four program locations, monthly average water temperature increased between 0.02 and 0.04°C per year. At one program location, monthly average water temperature decreased by 0.37°C per year. No detectable change in monthly average water temperature was observed at six locations. There was insufficient data to fit a model for four locations.

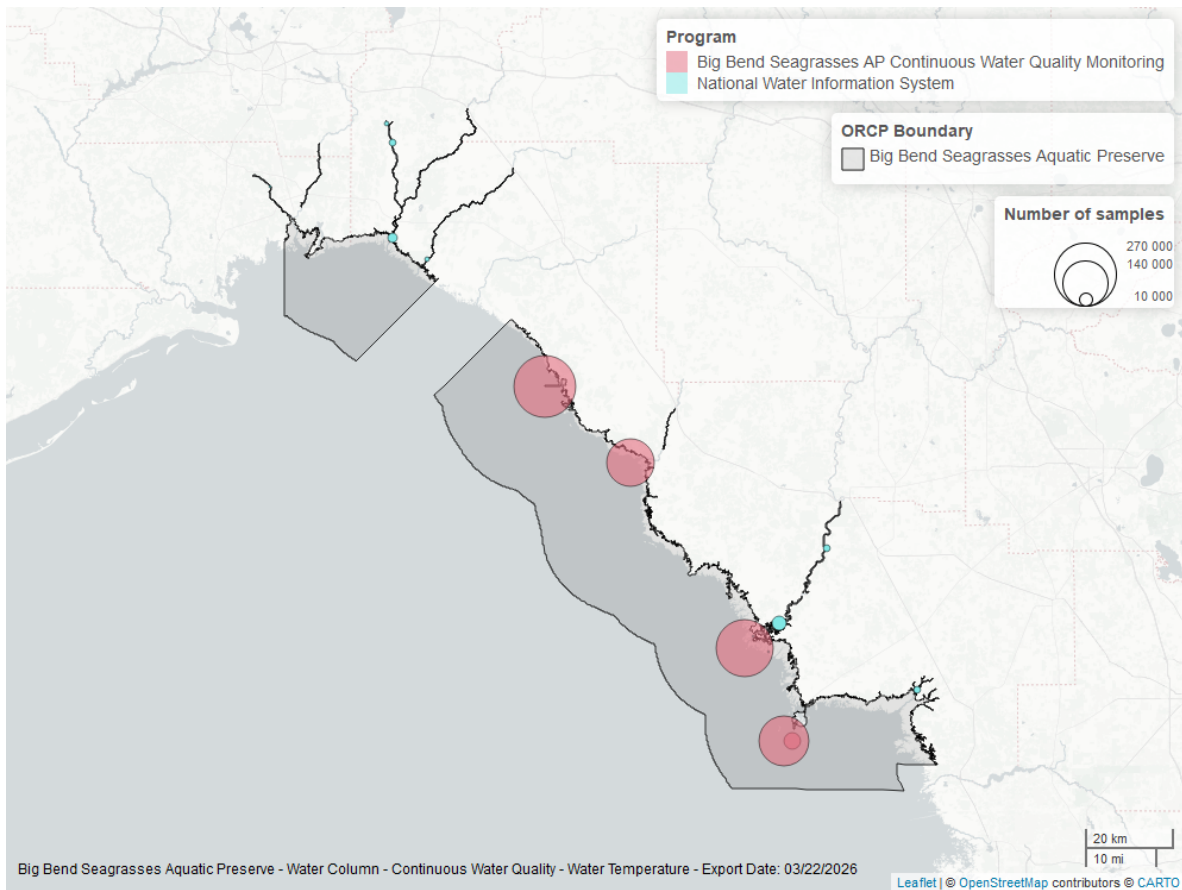


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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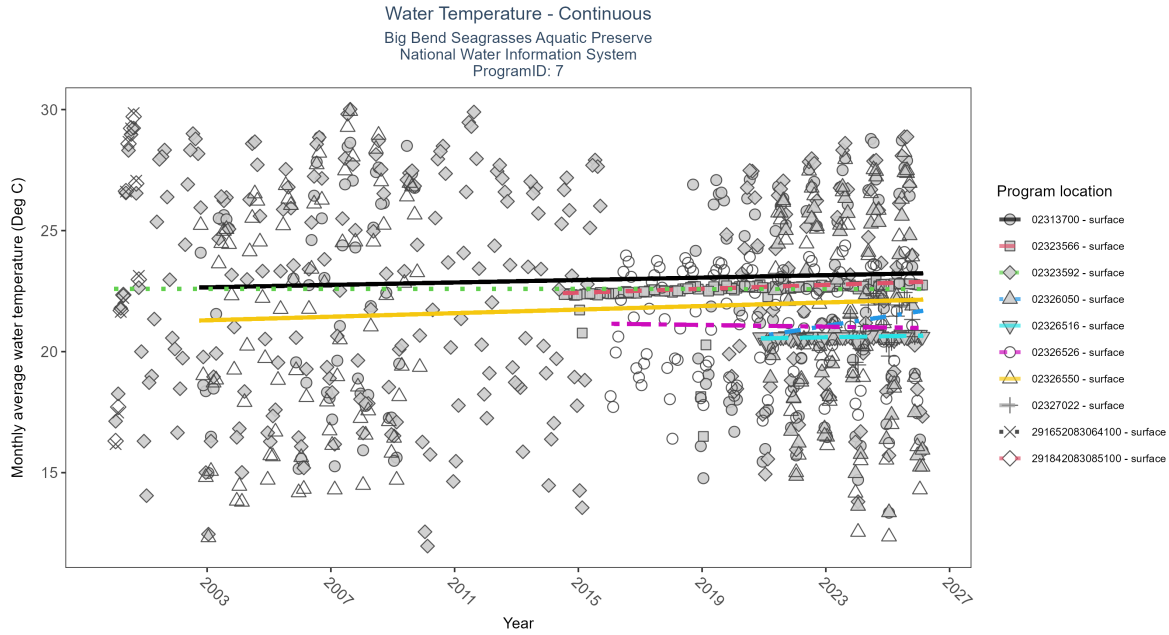


Figure 19: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: Seasonal Kendall-Tau Results for All Stations - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Significantly increasing trend	4275	16	2002 - 2026	22.9	0.14	22.63	0.03	0.0339
02323566	Significantly increasing trend	4200	13	2014 - 2026	22.6	0.78	22.39	0.04	0
02323592	No significant trend	13260	24	2000 - 2026	23.3	0	22.6	0	0.9807
02326050	No significant trend	1876	6	2021 - 2026	21.6	0.2	20.67	0.2	0.1039
02326516	Significantly increasing trend	1797	7	2020 - 2026	20.6	0.42	20.53	0.02	0.0002
02326526	No significant trend	3525	10	2016 - 2025	21.5	-0.08	21.15	-0.02	0.2898
02326550	Significantly increasing trend	5202	14	2002 - 2026	22.3	0.21	21.26	0.04	0.0013
02327022	Insufficient data to calculate trend	743	3	2023 - 2025	21.4	-	-	-	-
291652083064100	Insufficient data to calculate trend	473	1	2000 - 2000	26.1	-	-	-	-
291842083085100	Insufficient data to calculate trend	542	1	2000 - 2000	23.1	-	-	-	-

At four program locations, monthly average water temperature increased between 0.02 and 0.04°C per year. At one program location, monthly average water temperature decreased by 0.37°C per year. No detectable change in monthly average water temperature was observed at six locations. There was insufficient data to fit a model for four locations.

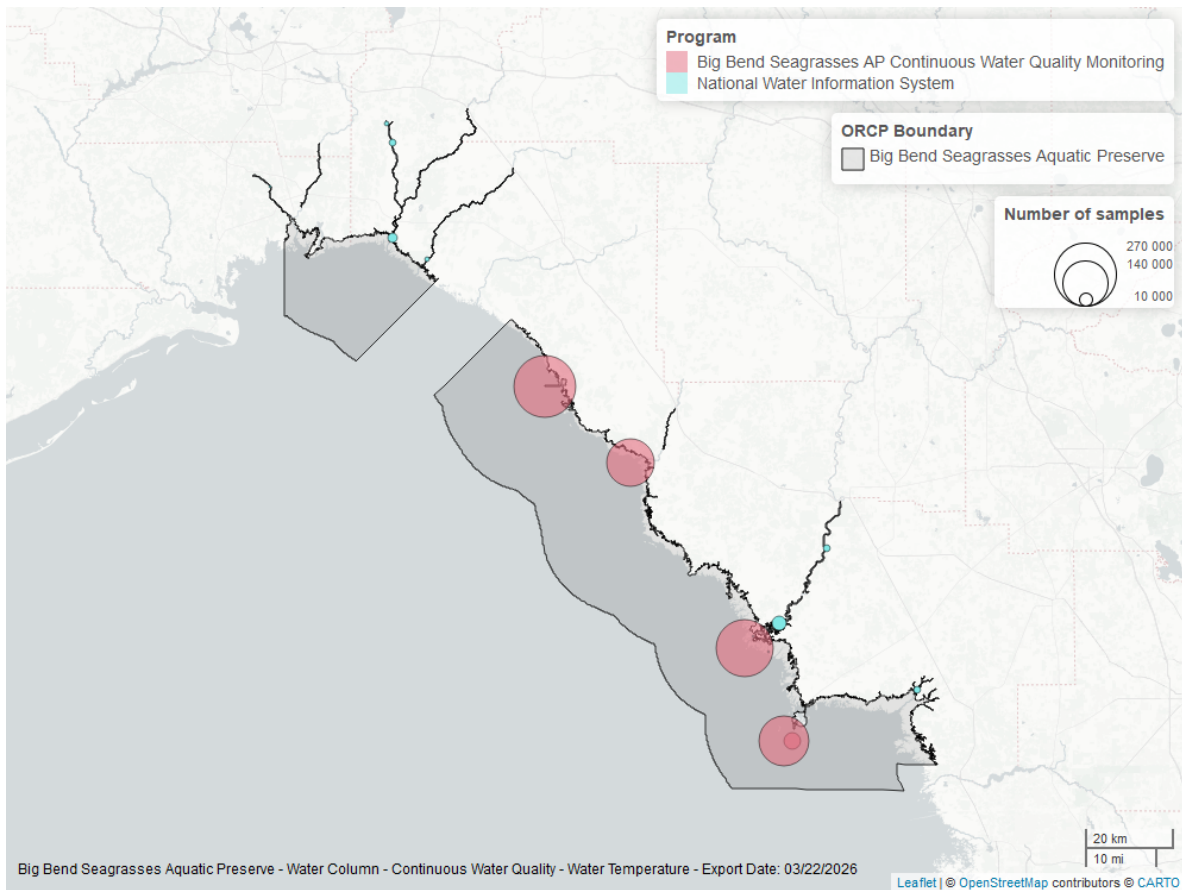


Figure 20: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

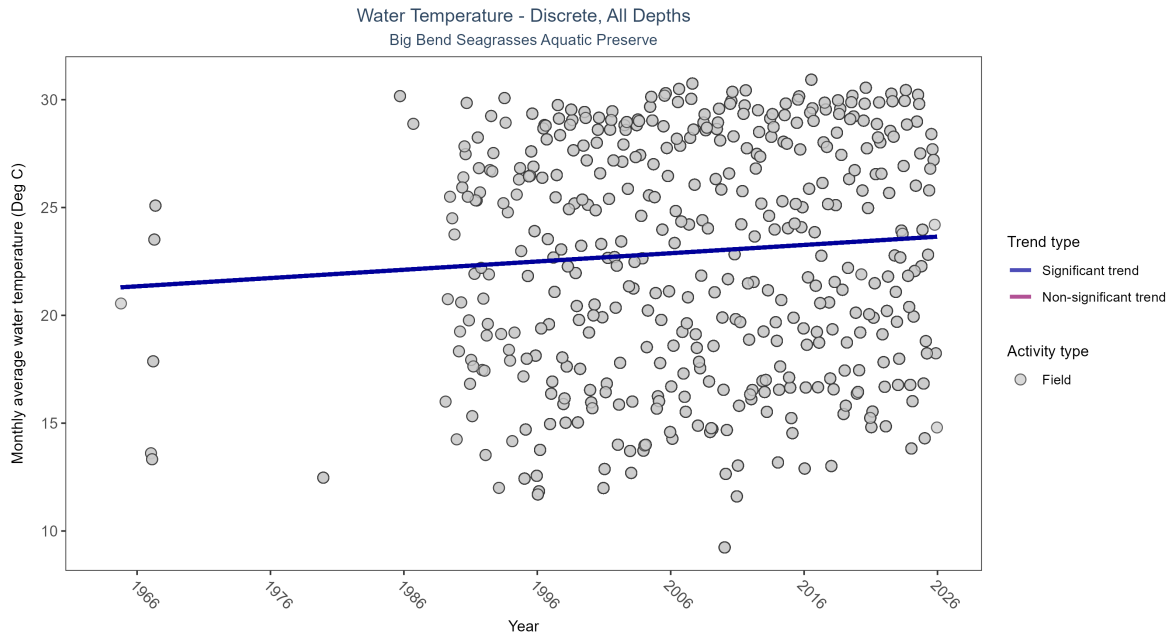


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 11: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	160088	42	1964 - 2025	24.1	0.21014	21.26901	0.03838	0

Monthly average water temperature increased by 0.04°C per year.

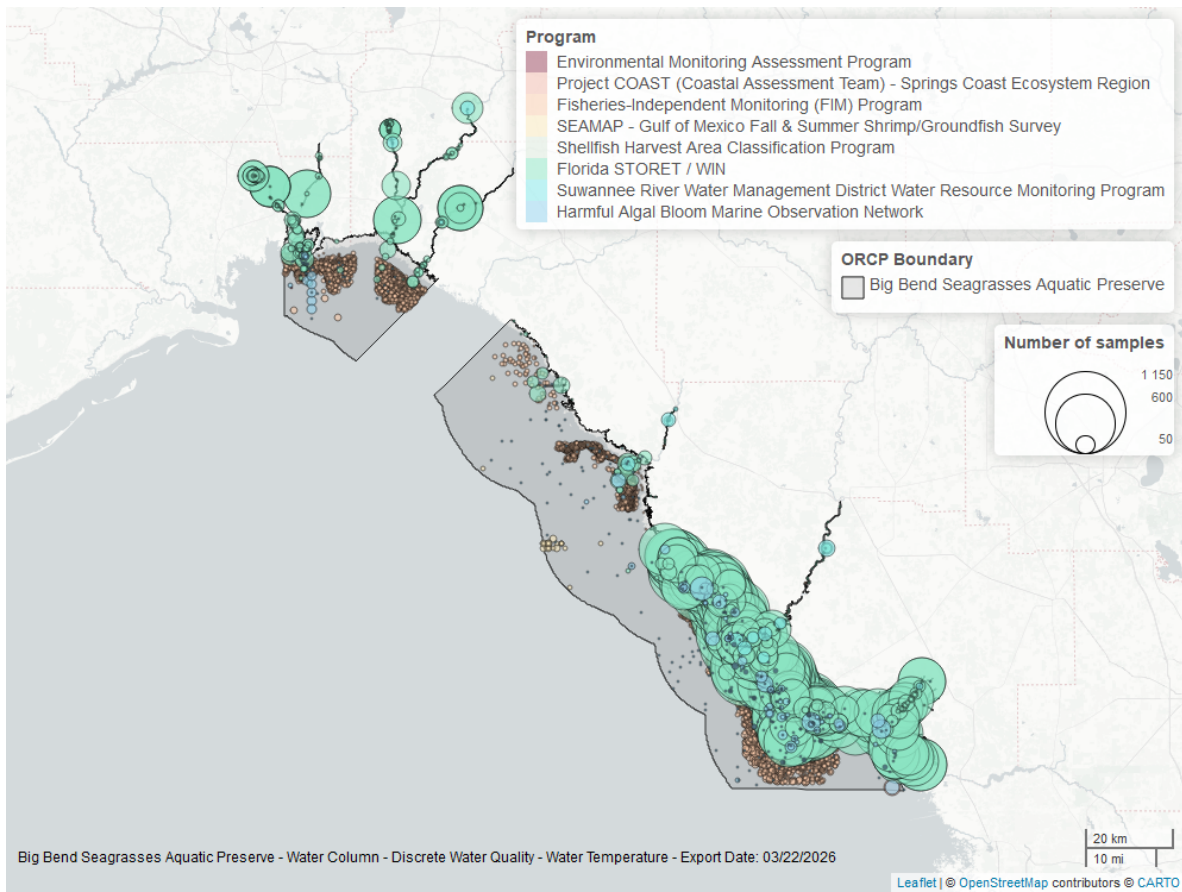


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

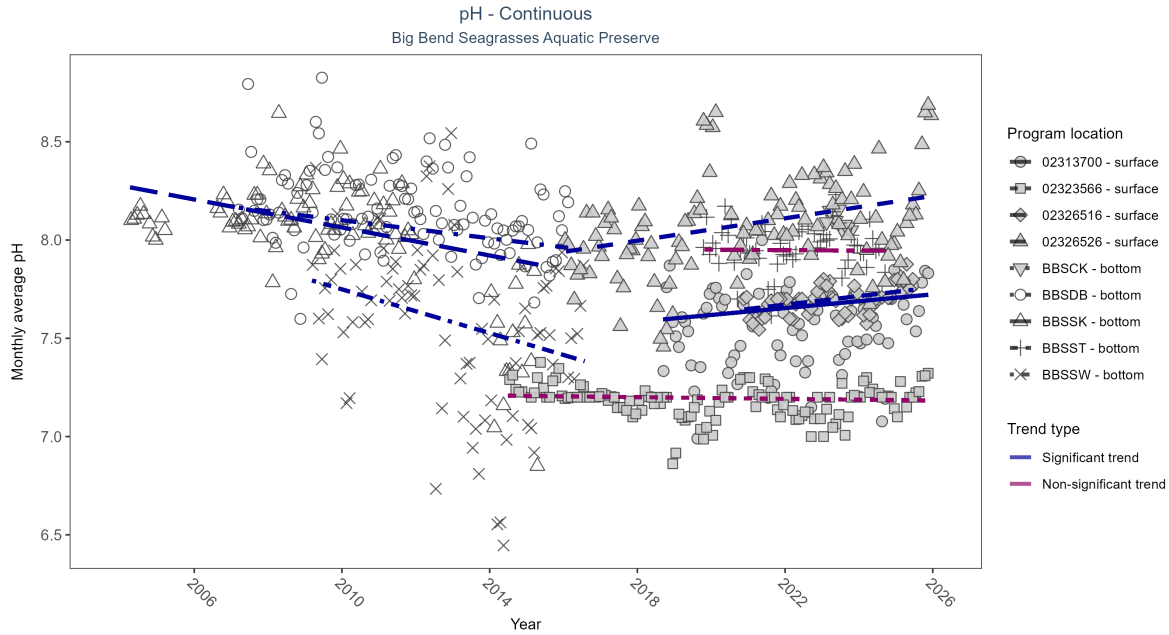


Figure 23: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Significantly increasing trend	2470	8	2018 - 2025	7.6	0.22	7.58	0.02	0.0222
02323566	No significant trend	3751	12	2014 - 2025	7.2	-0.08	7.21	0	0.2456
02326516	Significantly increasing trend	1485	6	2020 - 2025	7.7	0.4	7.63	0.02	0.0005
02326526	Significantly increasing trend	3274	10	2016 - 2025	8.0	0.26	7.94	0.03	0.0007
BBSCCK	Insufficient data to calculate trend	18185	1	2023 - 2023	8.1	-	-	-	-
BBSDDB	Significantly decreasing trend	250183	10	2007 - 2016	8.1	-0.28	8.17	-0.02	0.0004
BBSSK	Significantly decreasing trend	168278	10	2004 - 2015	8.1	-0.37	8.28	-0.04	0
BBSSST	No significant trend	153335	6	2019 - 2024	8.0	-0.01	7.95	0	0.9412
BBSSSW	Significantly decreasing trend	224733	8	2009 - 2016	7.6	-0.29	7.8	-0.06	0.0017

At three program locations, monthly average pH increased between 0.02 and 0.03 pH units per year. At three program locations, monthly average pH decreased between 0.02 and 0.06 pH units per year. No detectable change in monthly average pH was observed at two locations. There was insufficient data to fit a model for one location.

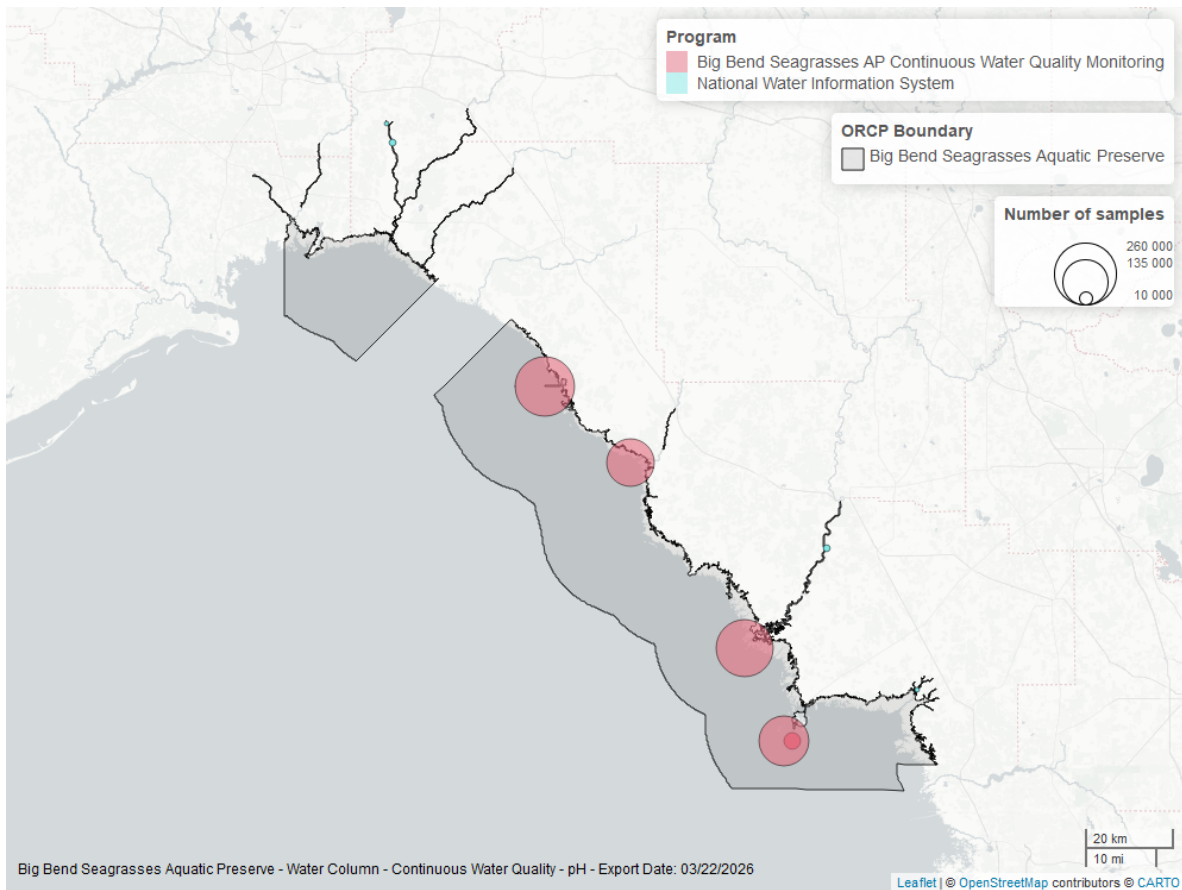


Figure 24: Map showing location of pH continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

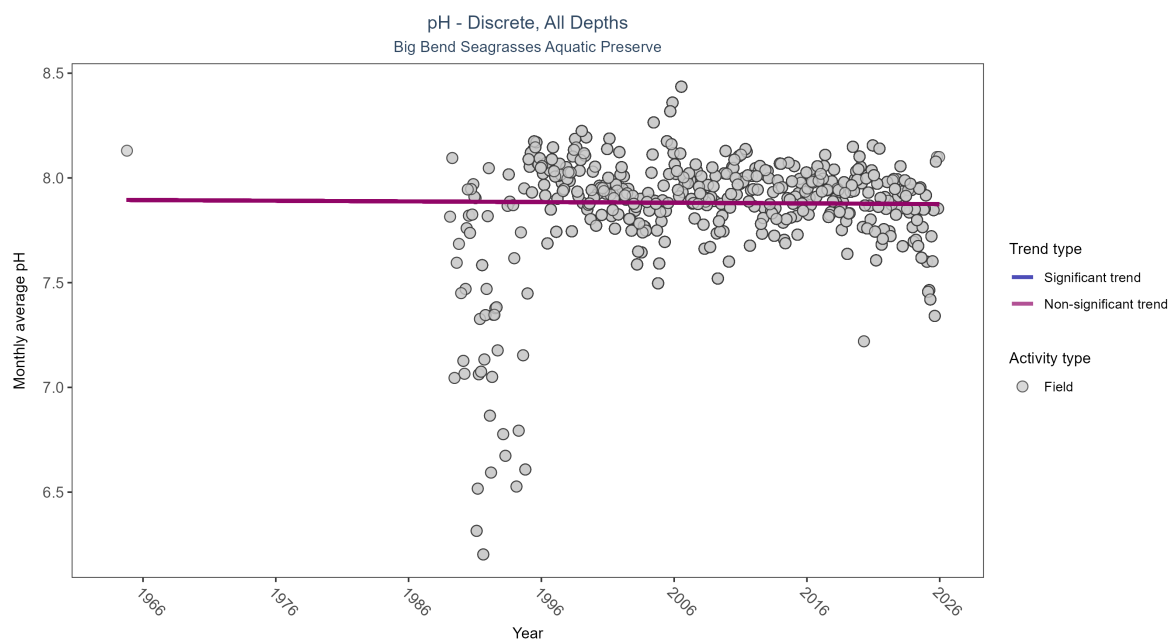


Figure 25: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 13: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	110666	38	1964 - 2025	8	-0.01356	7.89496	-0.00032	0.7846

pH showed no detectable trend between 1964 and 2025.

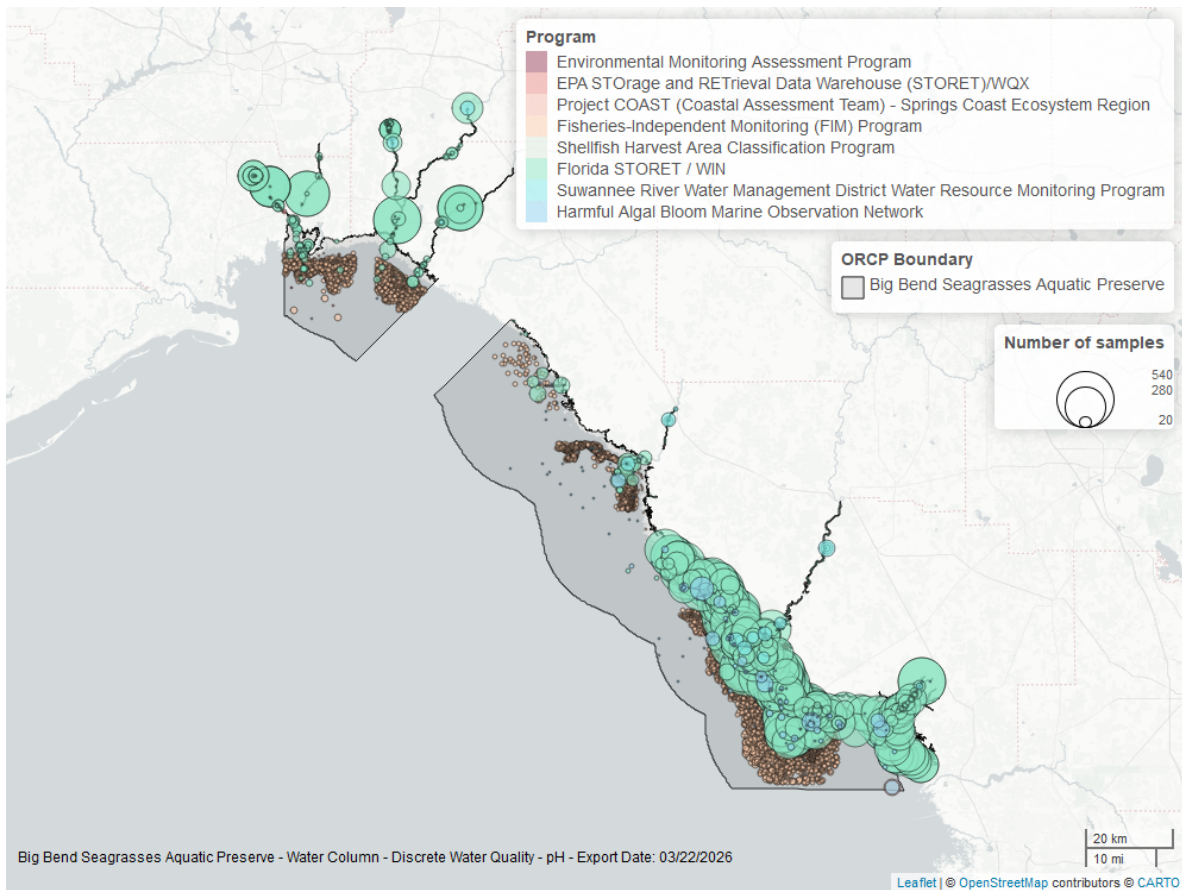


Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

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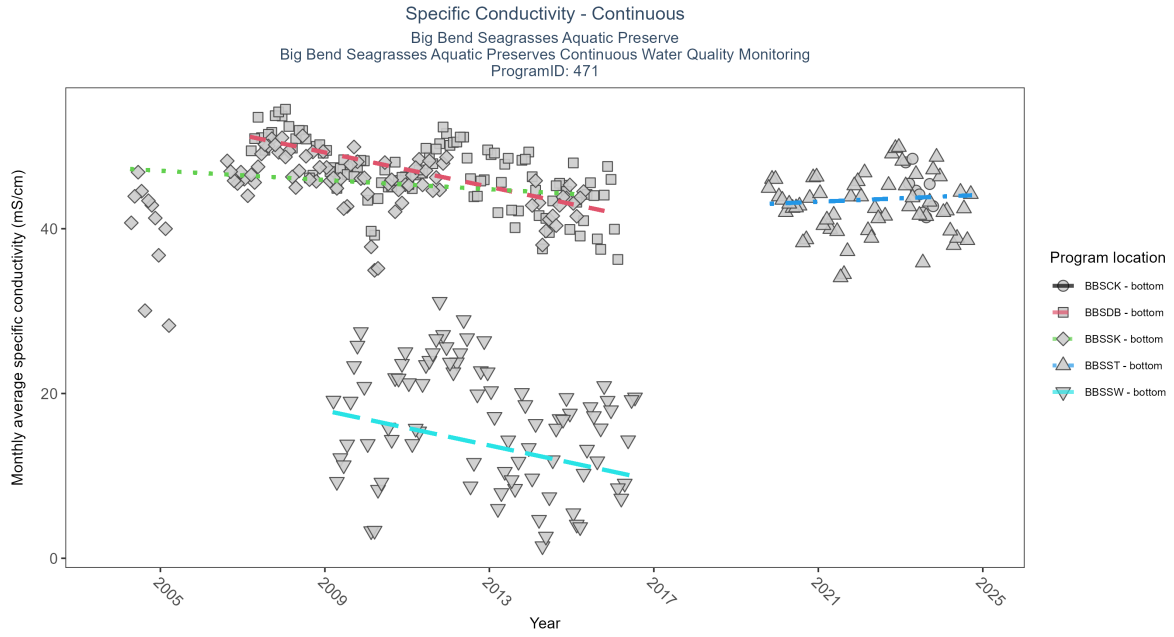


Table 14: Seasonal Kendall-Tau Results for All Stations - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSCCK	Insufficient data to calculate trend	14782	1	2023 - 2023	45.17	-	-	-	-
BBSDDB	Significantly decreasing trend	265988	10	2007 - 2016	47.08	-0.53	51.35	-1.04	0
BBSSK	Significantly decreasing trend	178359	10	2004 - 2015	45.80	-0.22	47.32	-0.28	0.0197
BBSSST	No significant trend	158160	6	2019 - 2024	44.13	0.08	42.86	0.21	0.516
BBSSSW	Significantly decreasing trend	221696	8	2009 - 2016	13.01	-0.23	17.94	-1.06	0.016

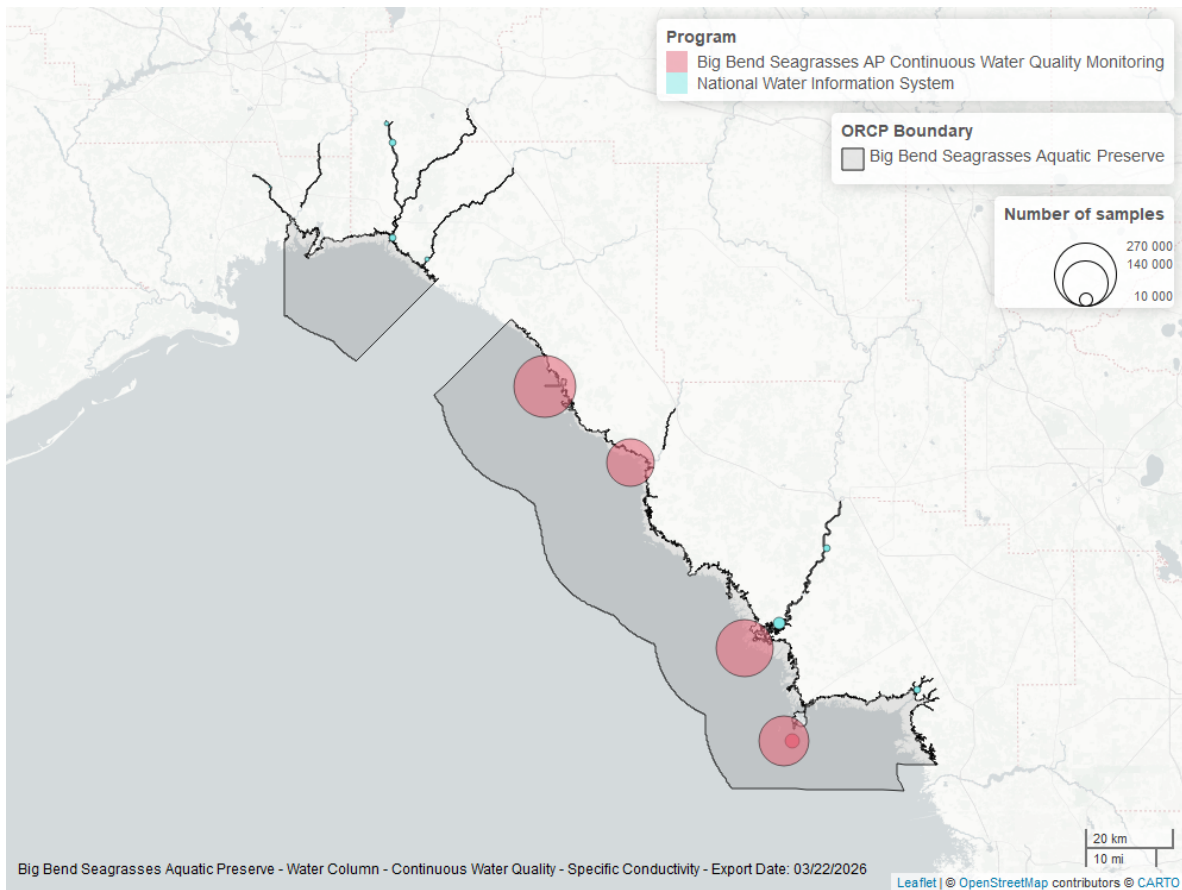


Figure 27: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

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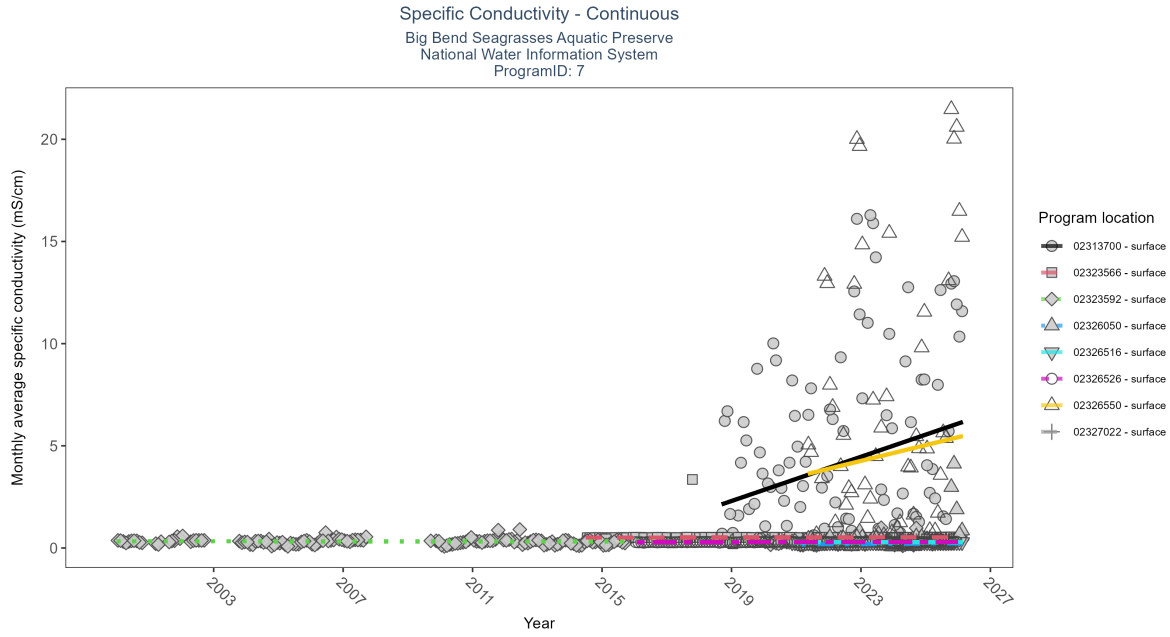


Table 15: Seasonal Kendall-Tau Results for All Stations - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02313700	Significantly increasing trend	2604	9	2018 - 2026	3.43	0.32	1.76	0.54	0.0003
02323566	Significantly increasing trend	4162	13	2014 - 2026	0.52	0.46	0.51	0	0
02323592	No significant trend	10826	23	2000 - 2026	0.31	-0.06	0.33	0	0.216
02326050	No significant trend	2166	6	2021 - 2026	0.24	0.12	0.17	0.01	0.358
02326516	Significantly increasing trend	1730	7	2020 - 2026	0.28	0.36	0.28	0	0.0012
02326526	Significantly increasing trend	3511	10	2016 - 2025	0.29	0.23	0.29	0	0.0015
02326550	No significant trend	3071	6	2021 - 2026	2.49	0.14	3.49	0.39	0.2688
02327022	Insufficient data to calculate trend	694	3	2023 - 2025	0.32	-	-	-	-

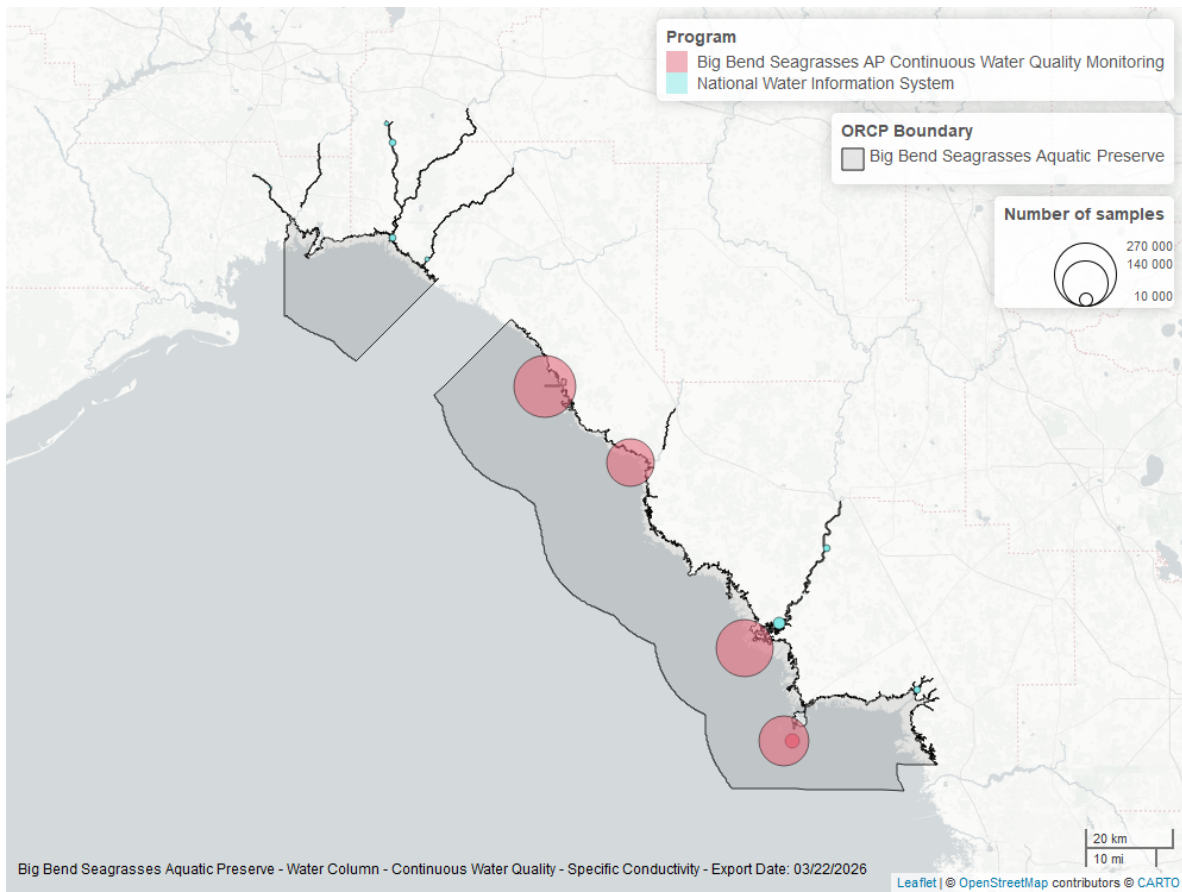


Figure 28: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

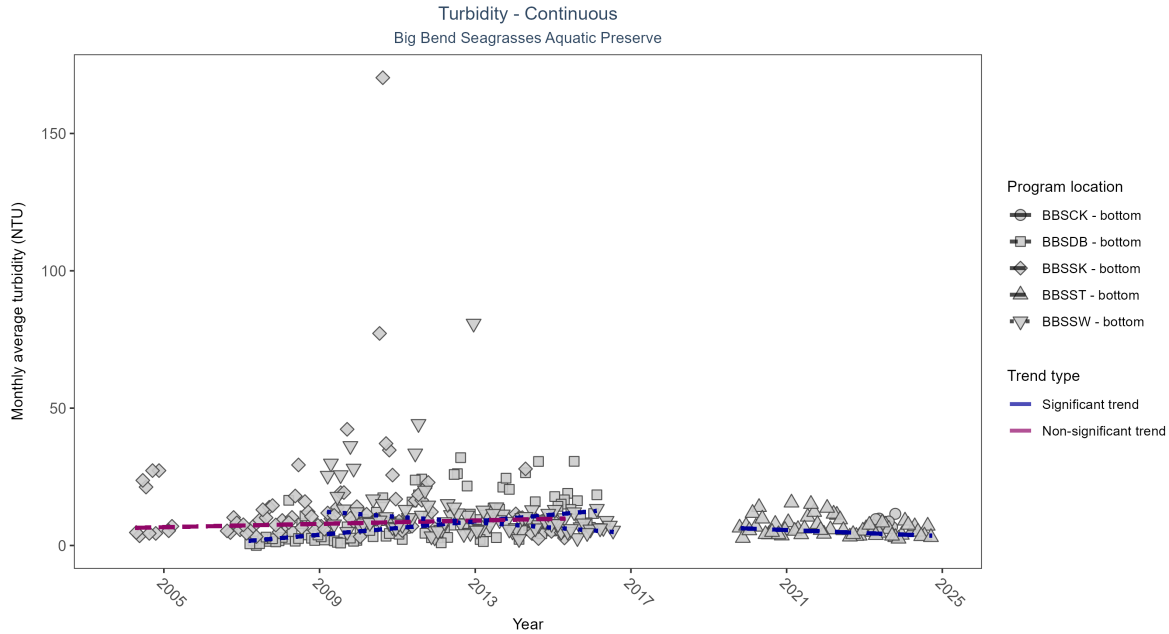


Figure 29: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 16: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
BBSC	Insufficient data to calculate trend	23243	1	2023 - 2023	7	-	-	-	-
BBSD	Significantly increasing trend	224613	10	2007 - 2016	1	0.49	1.47	1.22	0
BBSS	No significant trend	165043	10	2004 - 2015	5	0.11	6.35	0.3	0.1867
BBST	Significantly decreasing trend	157393	6	2019 - 2024	4	-0.31	6.73	-0.55	0.0115
BBSSW	Significantly decreasing trend	202699	8	2009 - 2016	6	-0.35	12.41	-0.99	0.0001

At one program location, monthly average turbidity increased by 1.22 NTU per year. At two program locations, monthly average turbidity decreased by 0.55 NTU per year at one site and by 0.99 NTU per year at the other. No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for one location.

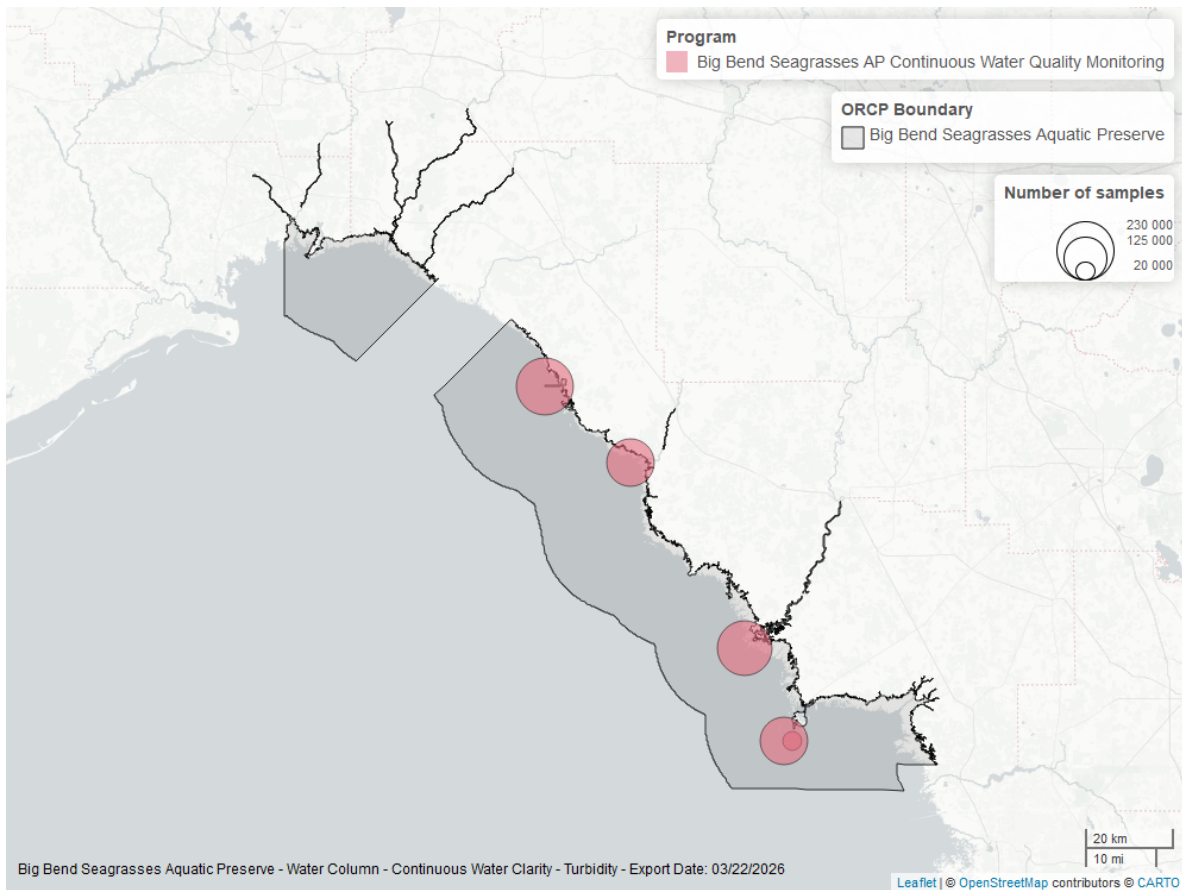


Figure 30: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

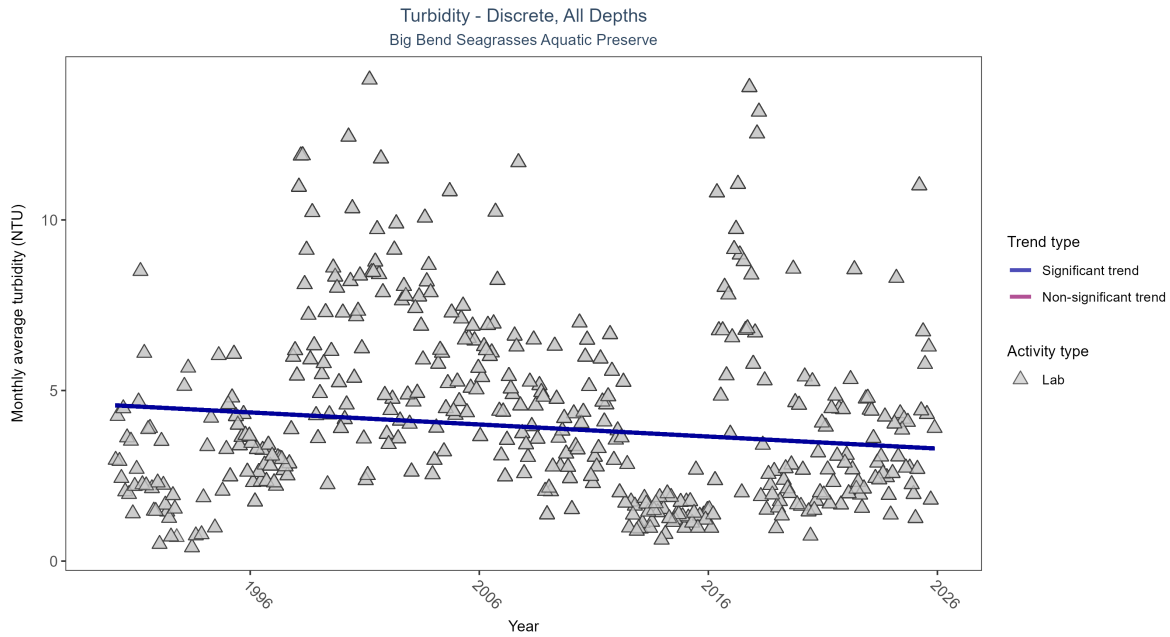


Figure 31: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	43028	36	1990 - 2025	3.4	-0.10626	4.56937	-0.03533	0.002

Monthly average turbidity decreased by 0.04 NTU per year, indicating an increase in water clarity.

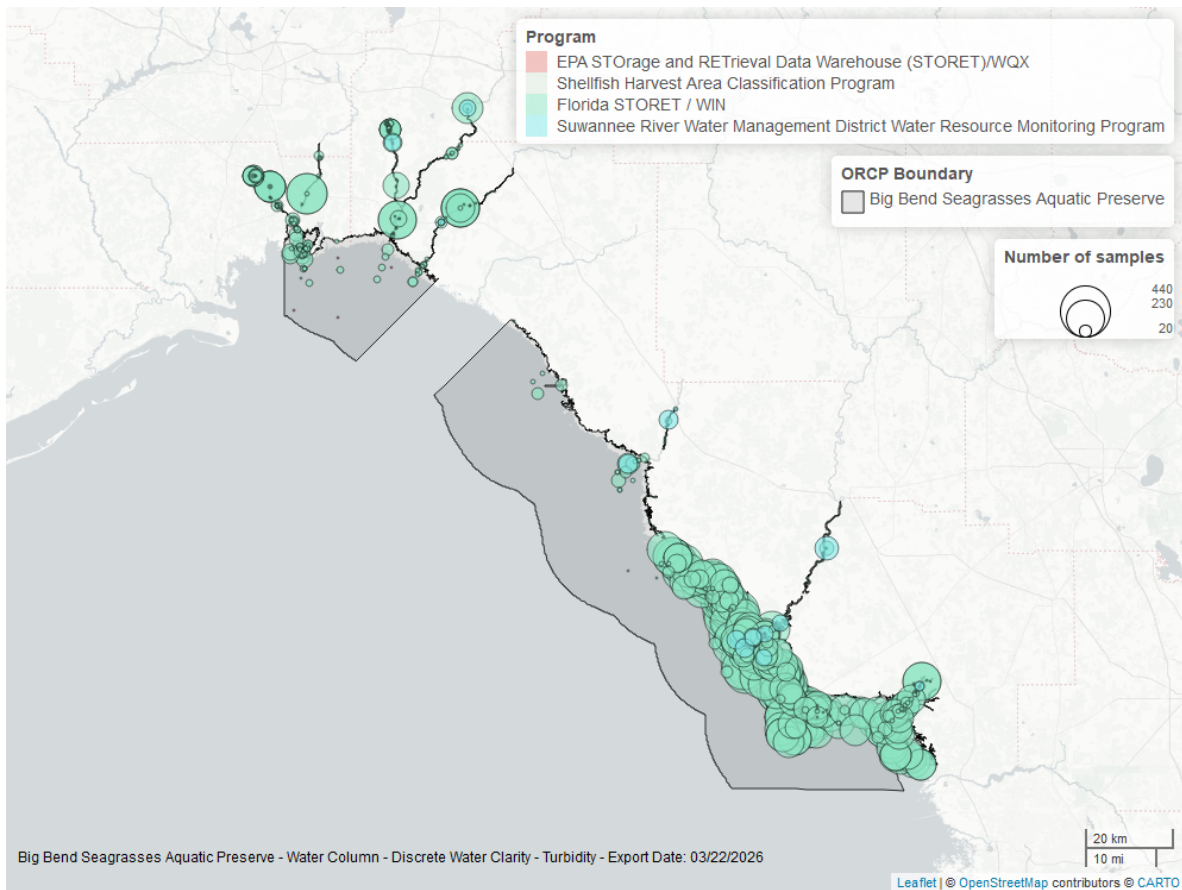


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

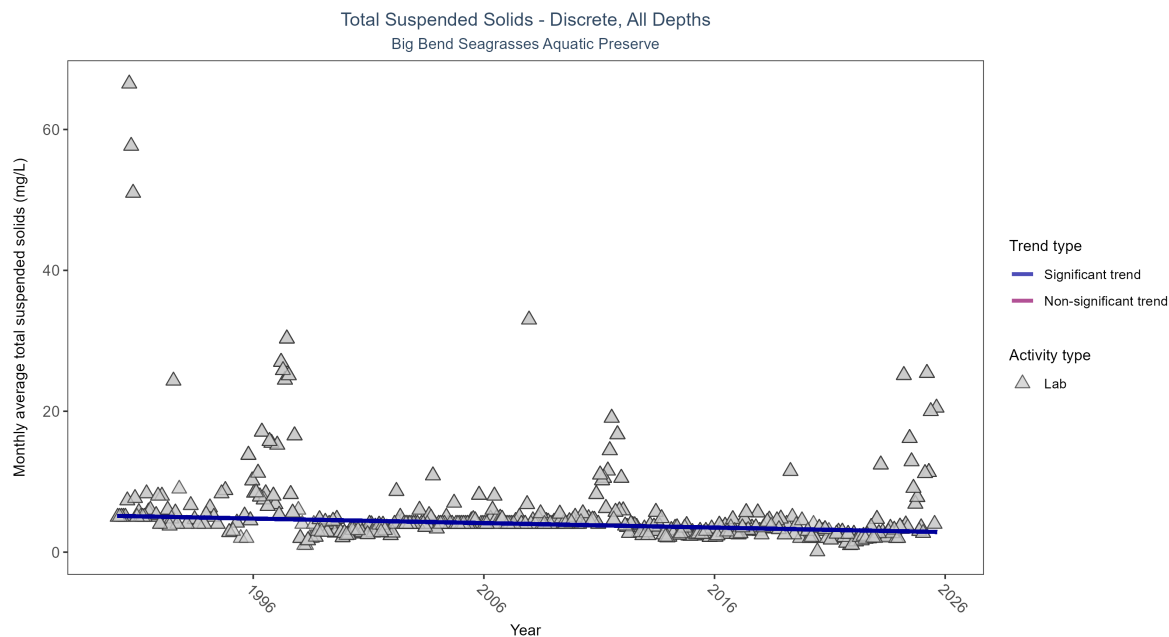


Figure 33: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	3036	36	1990 - 2025	4	-0.28684	5.15153	-0.06405	0

Monthly average total suspended solids decreased by 0.06 mg/L per year, indicating an increase in water clarity.

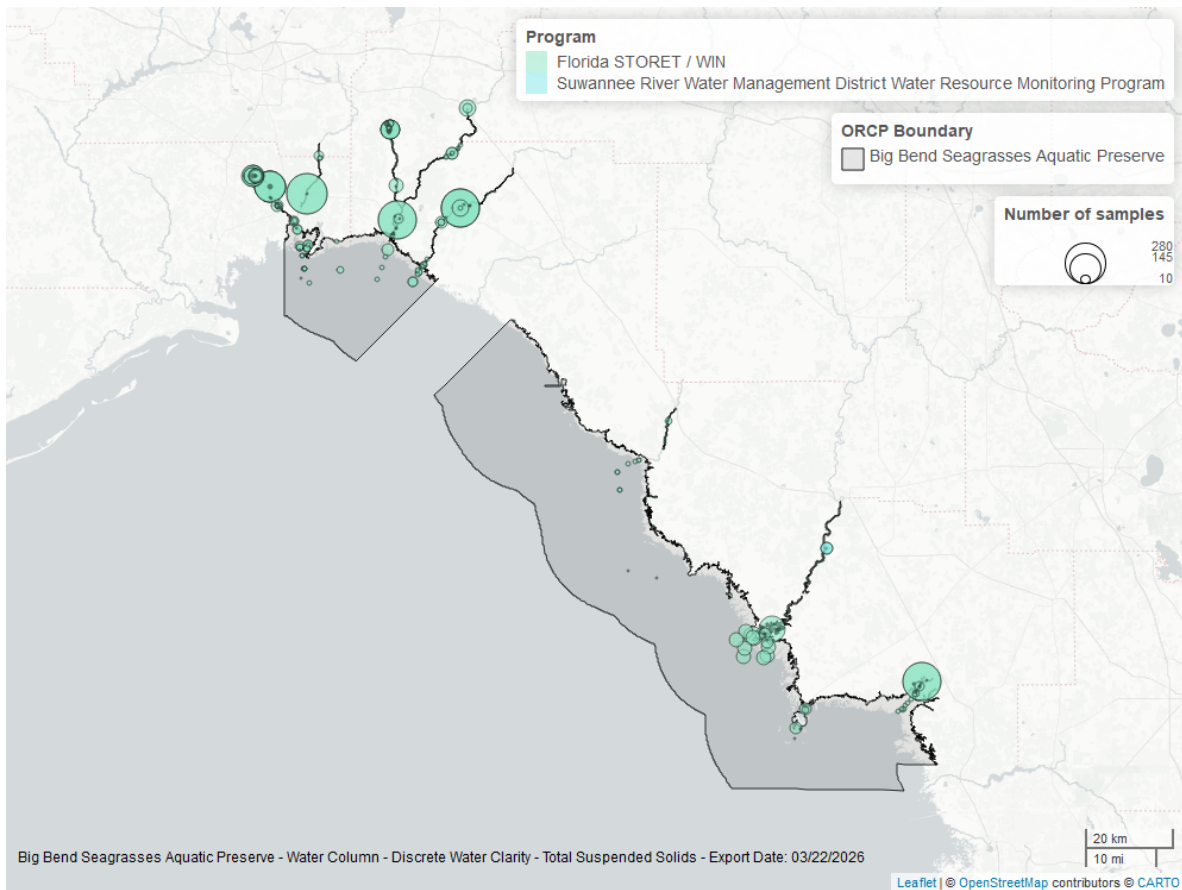


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

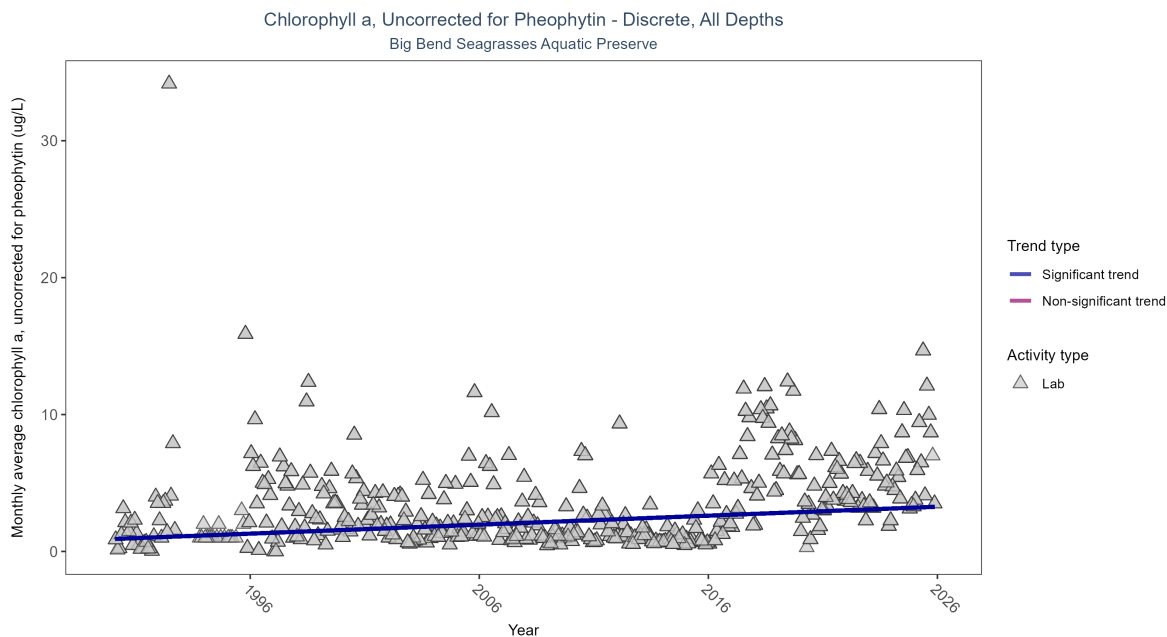


Figure 35: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 19: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	6422	36	1990 - 2025	1.2	0.23167	0.90671	0.06555	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by $0.07 \mu\text{g/L}$ per year, indicating a decrease in water clarity.

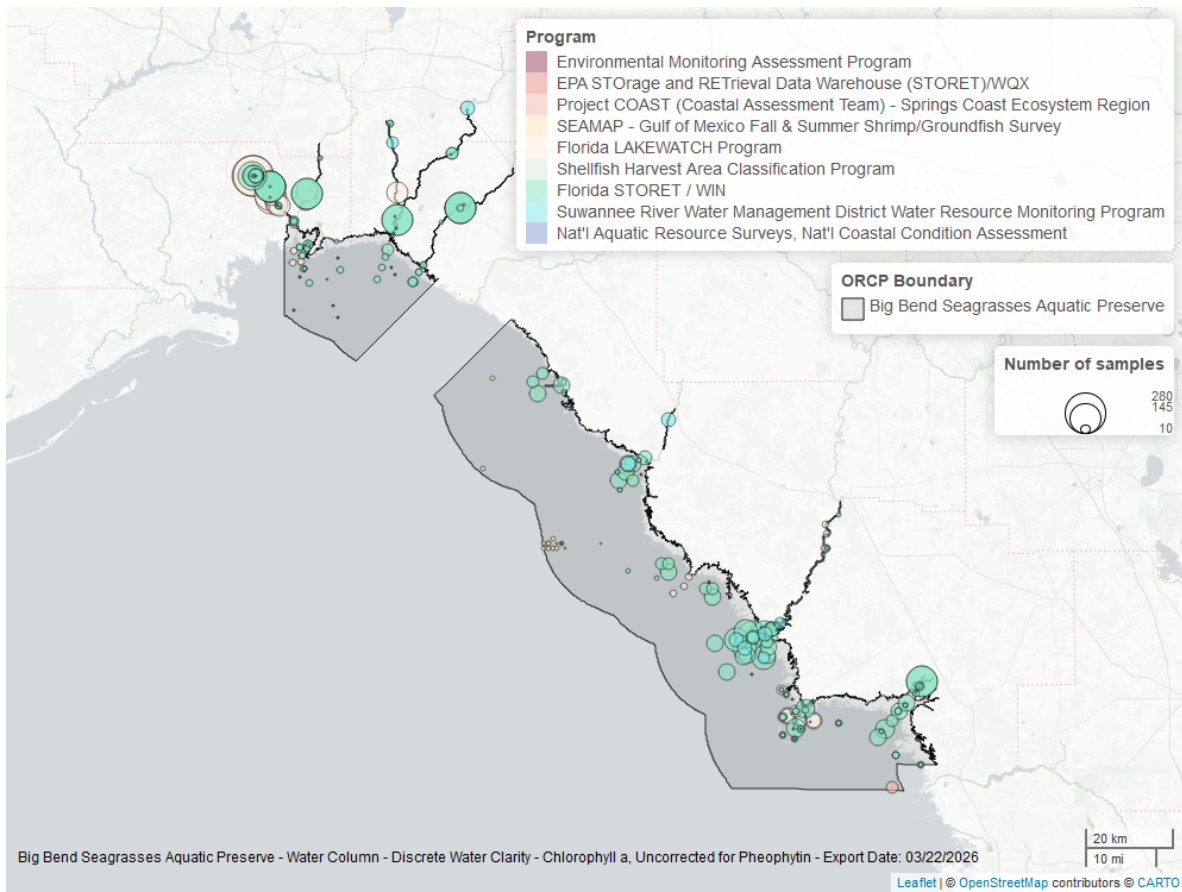


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

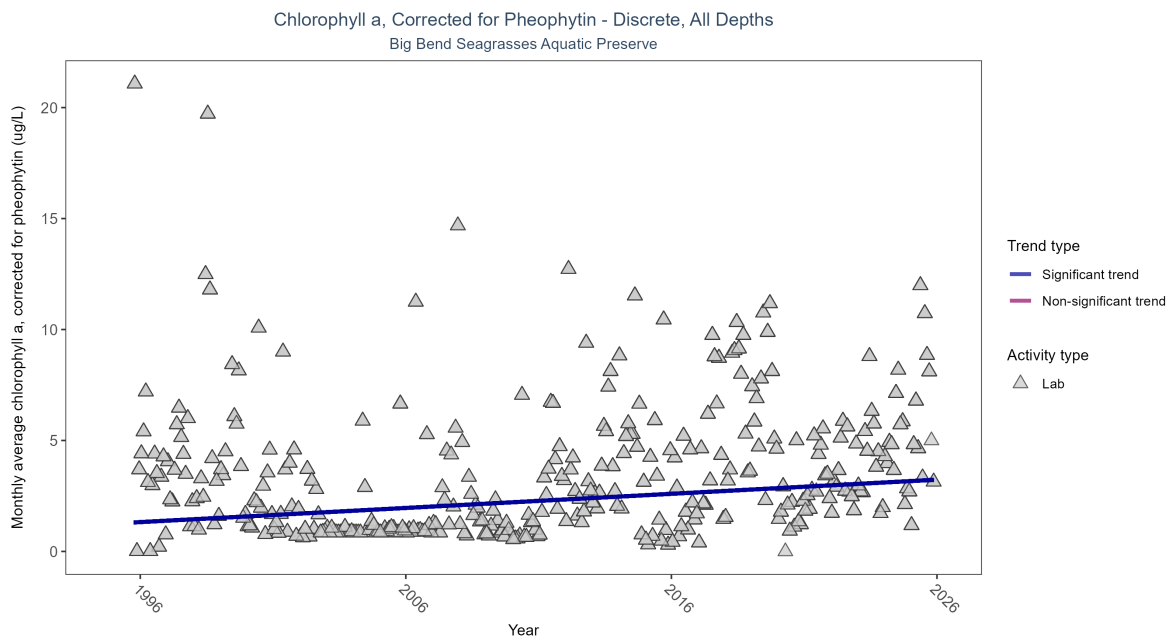


Figure 37: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	5053	31	1995 - 2025	1.1	0.19194	1.25854	0.06372	0

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.06 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

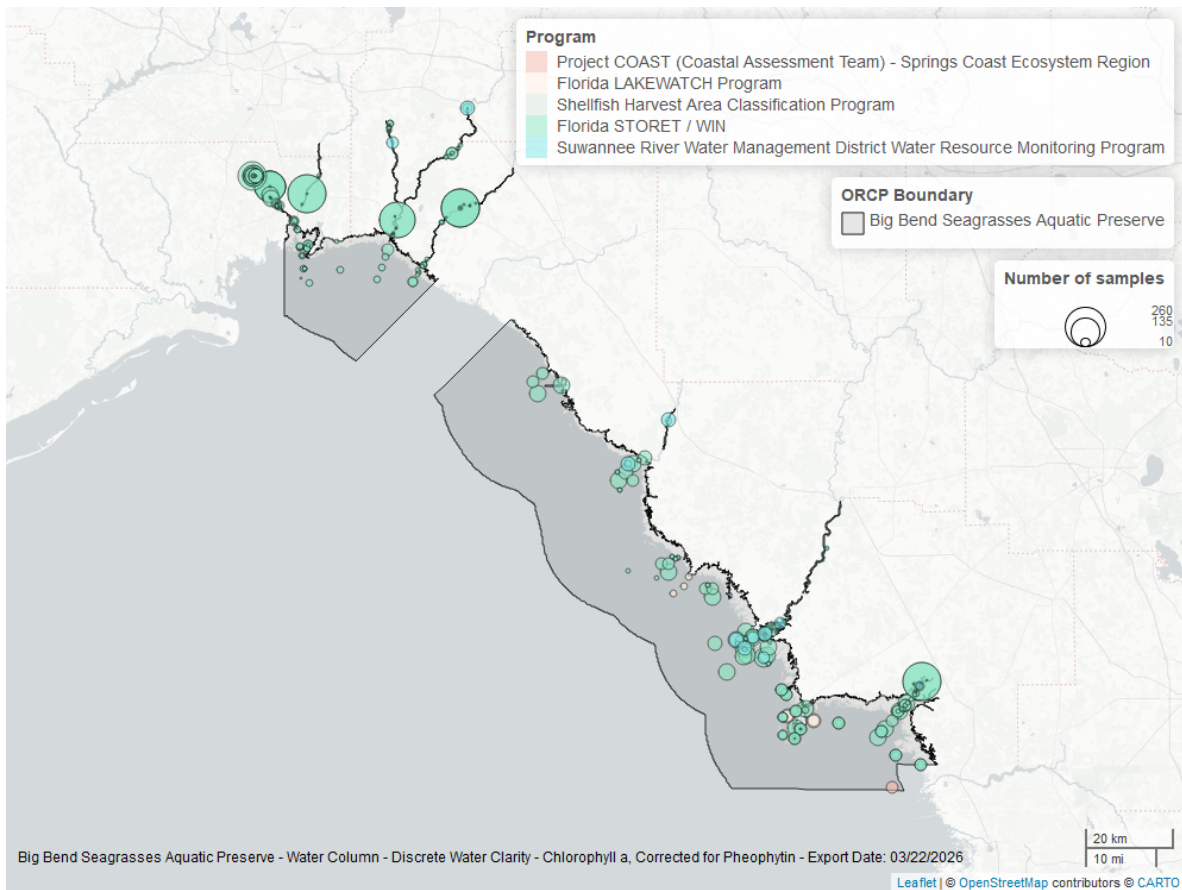


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

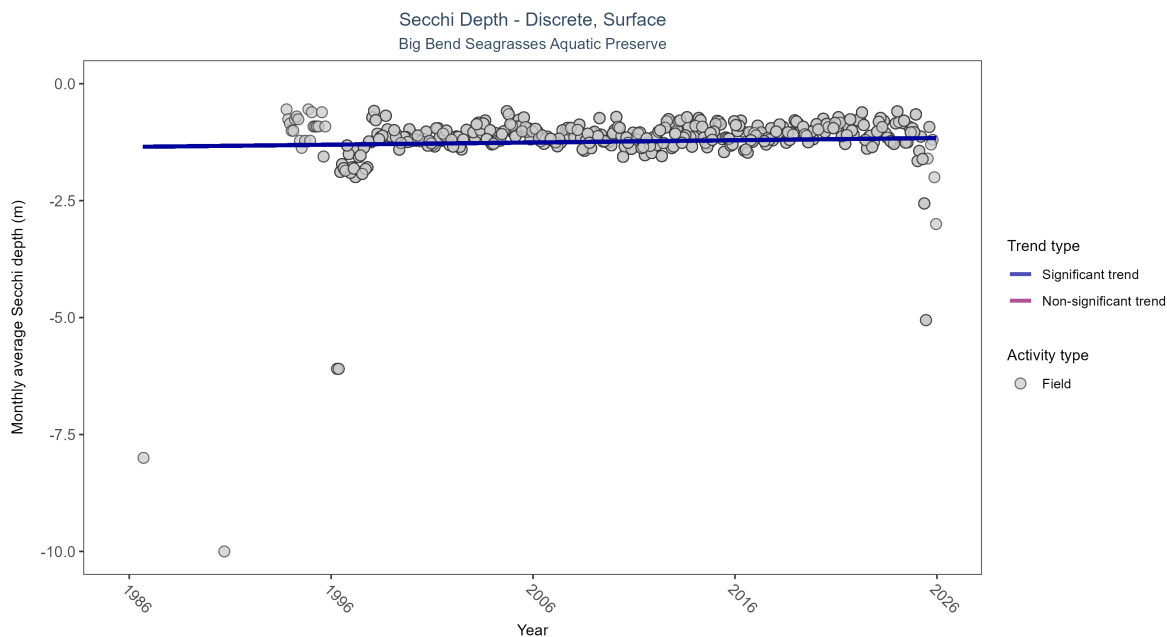


Figure 39: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 21: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	62821	37	1986 - 2025	-0.9	0.09869	-1.35154	0.00474	0.0066

Monthly average Secchi depth became shallower by less than 0.01 m per year, indicating a decrease in water clarity.

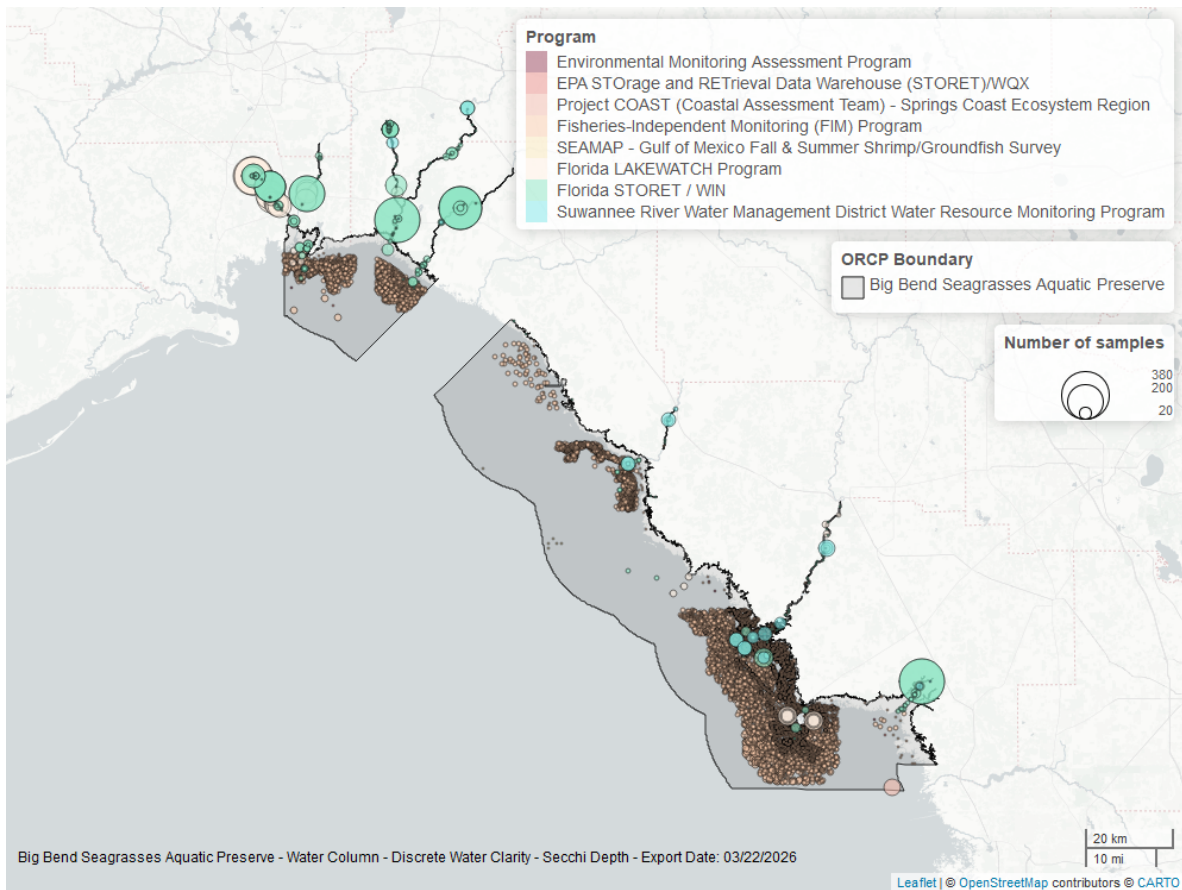


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

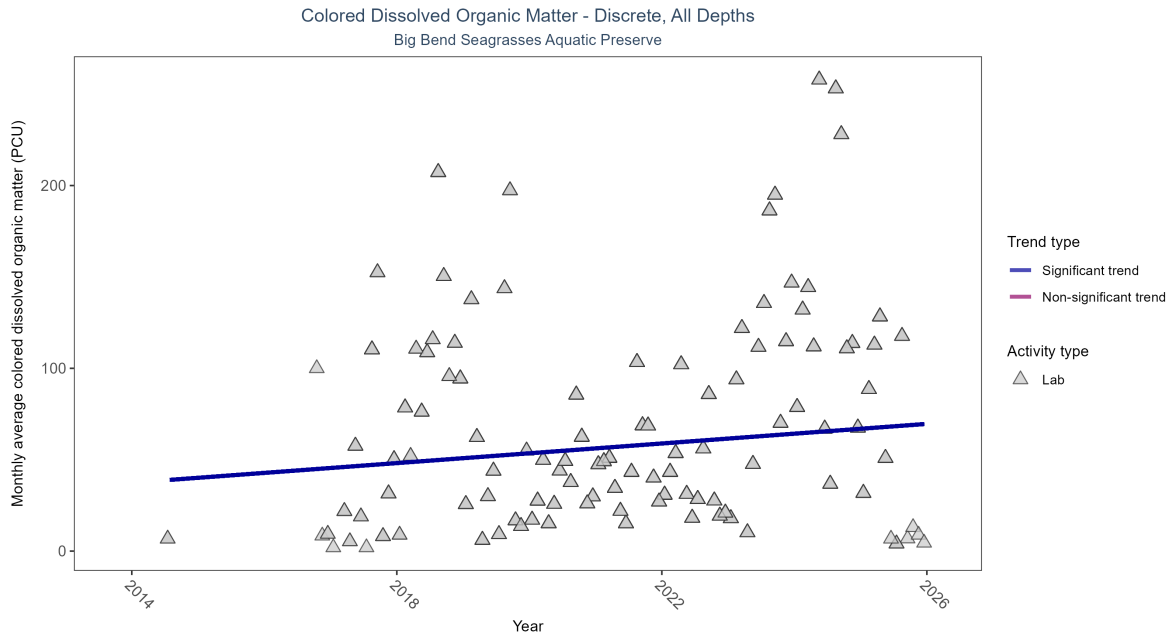


Figure 41: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 22: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	2186	11	2014 - 2025	22.8	0.17446	37.41909	2.68178	0.0268

Monthly average colored dissolved organic matter increased by 2.68 PCU per year, indicating a decrease in water clarity.

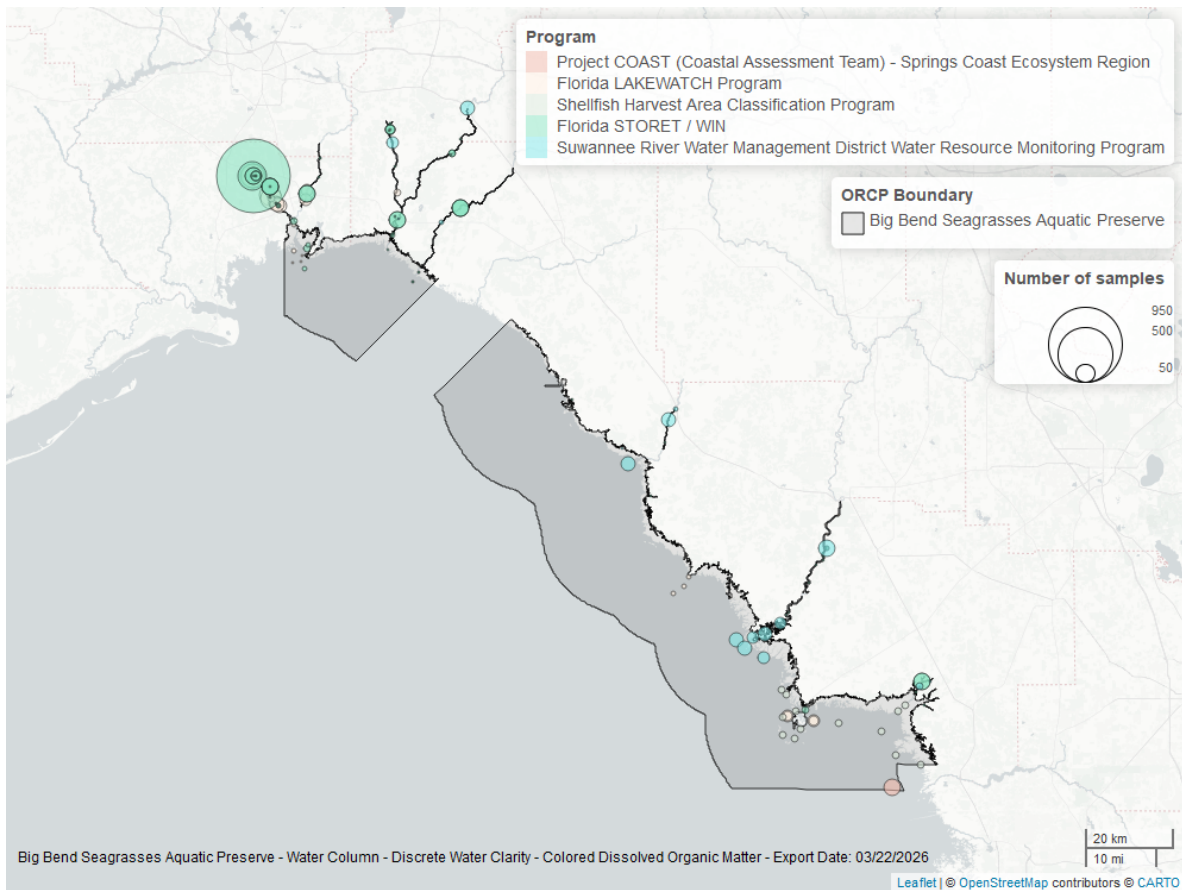


Figure 42: Map showing location of discrete water quality sampling locations within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.